

University of Sulaimani
College of Medicine

SPECIAL SURGERY BLOCK

INTRODUCTION

Welcome to the SS Block. We hope you get the principal yet essential ENT, ophthalmological & dermatological knowledge as well as communication, manual and some surgical skills.

Phase 2 Coordinator: Dr. Dana Ahmed Sharif / Prof. Nephrologist

Block Leader: Dr. Mariwan Latif Fatah /Ass.prof ENT

Co-leader Dr. Haval Ahmed Hama /Lecturer ENT

Ophthalmology coordinator: Dr. Tara Mahmood Hassan / Assis. prof/Ophthalmology

Plastic surgery coordinator: Dr. Suzan fazil / assis.prof/ plastic surgeon

Block Facilitators/ Teachers

ENT

Dr. Muaid Ismaeel Aziz,,,,,,,,,	Prof./ENT
Dr. Sherko Saeed	Prof./ENT
Dr. Mariwan Lateef Fattah	Ass. prof/ENT
Dr. Haval Ahmad Hama	Lecturer/ENT

Ophthalmology

Dr. Baxtear Qadir Hamasalih.	Ass.prof/Ophthalmology
Dr. Tara Mahmood Hasan	Ass. Prof./Ophthalmology
Dr. Khalid Saeed Abdulmajeed.	Ass. Prof./Ophthalmology
Dr. Deri Latif Amin.	Ophthalmology

Plastic surgery

Dr Ari Raheem Qader.....	Prof./ plastic surgery
Dr Jalal Ali	Ass,Prof./ plastic surgery
Dr Suzan fadhil Jaber. .	lecturer /plastic surgery
Dr Dana abdulmajeed.	Ass.prof/plastic surgery

AIMS OF THE BLOCK

- 1) Take a history and carry out an appropriate examination in a patient with an ENT, ophthalmic & dermatological problem.
- 2) Understand the impact of dysfunction or loss of a special sense for a patient and their career and the resources required to manage the disability.
- 3) Understand acute, common, and important ENT. ophthalmological and dermatological disorders, especially those that have systemic features or appear in other parts of the MBChB course.

LEARNING OUTCOMES

ENT:

- Demonstrate their ability to identify the important causes of nasal blockage, rhinitis, epistaxis, deafness, pain in the ear, pain in the throat, difficulty swallowing, swelling of the neck, facial pain, headache, & hoarseness of voice.
- by taking an appropriate history to reach a provisional diagnosis.
- Elicit selectively normal and common abnormal signs in the ears, nose and throat including the use of an otoscope and a tuning fork to test diagnostic hypotheses.
- Use investigations selectively to confirm diagnostic hypotheses.
- Formulate a simple management plan including an assessment of the need for referral.

OPHTHALMOLOGY

By the end of the block the student should be able to:

- Demonstrate their ability to identify the important causes for ocular discomfort, visual disturbance, a red eye, ocular discharge, & an abnormal pupil.
- Take an appropriate history to reach a provisional diagnosis.
- Elicit selectively, normal and common abnormal signs in the eyes to test diagnostic hypotheses, in particular:
 - o test and record visual acuity in adults and children.
 - o perform the swinging flash light test for a relative afferent pupillary defect.
 - o assess a patient for the presence of squint by means of the corneal reflexes and cover testing.
 - o examine the anterior segment and external eye with a pen torch, examine the fundus with a direct ophthalmoscope.
 - o use mydriatic and fluorescein diagnostic drops safely.
 - o examine visual fields by confrontation.
 - o examine the ocular media of both adults and children by means of the red reflex.
- Distinguish between ophthalmic complaints requiring immediate referral,

those which require referral but are not urgent and those which can be managed by the newly qualified medical practitioner

- Discuss the extent and causes of preventable blindness world-wide

Plastic surgery

Objectives

- 1.The basics of wound healing,stages, and problems that lead to delay in wound healing.**
- 2.Reconstructive ladder and its applications in plastic surgery.**
- 3.The concept of body angiosomes .**
- 4.Skin grafts ,Biology,classification, indications and causes of falilure.**
- 5.Flaps classification,indications, and surgical techniques.**
- 6.Skin lesions (Benign & Malignant).**
- 7.Scars (Hypertrophic & keloid)**
- 8.Types of suturing & suture materials**
- 9.Burn types, management and indications for admission.**

BLOCK INFRASTRUCTURE

- It involves 6 weeks.
 - **First 2 weeks induction phase (only theory lecture)**
 - Start from Sunday to Thursday
 - From 8:00 am – 12:00pm.
 - Remaining 4 weeks practical lectures;
 - Place for practical lectures are:
 1. ENT center.
 2. Shaheed Aso Hospital.
 3. Burn Hospital.
- 2 weeks for ENT, 1 week for plastic surgery, 1 week for ophthalmology

INDUCTION PHASE LECTURES

Week 1	8- 9am	9-10 am	10-11 am	11-12 am
Sunday	Acute sinusitis Dr.Haval	Complication of acute sinusitis Dr.Haval	Neck mass Dr. Mariwan	Cataract and glaucoma Dr. Khalid
Monday	Orbit eyelid & lacrimal system Tropical ophthalmology Dr.Tara	Ocular emergency & trauma Dr.Tara	Peripheral auditory disorder Dr. Sherko	Peripheral auditory disorder Dr. Sherko
Tuesday	Allergic rhinitis and non-allergic rhinitis Dr. Haval	Sinonasal tumor Dr. Muaid	Cutaneous vascular malformation and Dr. sozan	Reconstructive ladder Dr. sozan
Wednesday	Chronic sinusitis Dr. Haval	Nasal polyposis Dr. Haval	Flap classification uses in plastic surgery Dr. jalal	Graft Dr. Jalal
Thursday	Scar and it is management Dr. Ari	Burn and it is management Dr. Ari	Introduction, anatomy & clinical Optics Dr. Bakhtyar	Ocular surface disease Dr. Bakhtyar

INDUCTION PHASE LECTURES

Week 2	8- 9am	9-10 am	10-11 am	11-12 am
Sunday	Otitis externa Dr. Muaid	Acute otitis media and OME Dr. Muaid	Epistaxis Dr. Haval	Nasal trauma Dr. Haval
Monday	Peripheral vestibular disorder Dr. Sherko	Peripheral vestibular disorder Dr. Sherko	Squint and peditric ophthalmology Dr. Dere	Retina and retinal vascular disease Dr. Dere
Tuesday	Chronic suppurative otitis media Dr. Muaid	Neck trauma Dr. Mariwan	Acute laryngitis Dr. Mariwan	chronic laryngitis Dr. Mariwan
Wednesday	Laryngeal tumor Dr. Mariwan	Laryngeal tumor Dr. Mariwan	Stridor and tracheostomy Dr. Mariwan	Uveitis and neurophthalmology Dr. Khalid
Thursday	Cleft lip and palate Dr. Dana	Skin lesions Dr. Dana	Acute and chronic pharyngitis Dr. Muaid	Pharyngeal tumor Dr. Mariwan

Lectures:

ENT:

EAR: each lectures.

1. Peripheral Auditory disorders [Dr. Sherko](#)
2. Peripheral vestibular disorders [Dr. Sherko](#)
3. Otitis externa . [Dr. Muaid](#)
4. Acute otitis media and media with effusion. [Dr. Muaid](#)
5. C.S.O.M [Dr. Muaid](#)

NOSE and PNS : each lectures 2 hours

1. Allergic rhinitis and nasal polyposis. [Dr. Haval](#)
2. Epistaxis and nasal trauma. [Dr. Haval](#)
3. Acute Rhinosinusitis and it is Complication. [Dr. Haval](#)
4. Chronic rhinosinusitis and allergic rhinitis. [Dr. Haval](#)

5. Sinonasal malignancy: Dr. Muaid

NECK ; each lectures 2 hours

- 1. Neck mass. Dr. Mariwan**
- 2. Neck trauma. Dr. Mariwan**

Pharynx: each lectures 2 hours

- 1. Acute and chronic pharyngitis. Dr. Sherko**
- 2. Pharyngeal tumors. Dr. Sherko**

Larynx; each lectures 2 hours

- 1. Acute and chronic laryngitis. Dr. Mariwan**
- 2. Laryngeal tumors. Dr. Mariwan**
- 3. Stridor and tracheostomy: Dr. Mariwan**

OPHTHALMOLOGY:

- **Dr Bakhtiar**

Introduction, anatomy & clinical Optics
Ocular surface disease

- **Dr Tara**

Orbit eyelid & lacrimal system
Tropical ophthalmology, ocular emergency & trauma

- **D.Dere**

Retina and retinal vascular disease
Squint and paediatric ophthalmology

- **D. Khalid**

Cataract and glaucoma
Uveitis and neurophthalmology

Plastic surgery :

- **Dr.Ari Rheem**

1. Scar management
2. Management of burn

- **Dr Suzan Fadhil**

1. Cutaneous vascular malformation
2. Reconstructive ladder

- **Dr. Dana**

1. Cleft lip and palate
2. Skin tumours (Benign & malignant)

- **Dr. Jalal Ali**

1. Flaps
classification and
uses in plastic
surgery.
2. Graft

PART TWO: POSTINDUCTION CLINICAL COURSE

Practical Session

- Session the students divided in to two groups A (A1, A2) and B (B1, B2) group A start practical sessions in ENT department for 2 weeks , group B start practical sessions in ophthalmology and plastic surgery each for one week after 2 weeks group A and group B exchanges

CLINICAL SESSIONS

History taking, Examination, Endoscopic and Microscopic Examination & Discussion

Diseases of ear, Approach to deafness & Surgery of middle ear

Diseases of Nose and PNS , Nasal obstruction and Sinonasal surgery

Diseases of Larynx and Pharynx, Dysphagia & Upper airway obstruction

Ear radiology

Nose radiology

Throat & neck radiology

Pharmacology in ENT

OPHTHALMOLOGY and plastic surgery

Operative theatre

Ward

Outpatient clinic&

Optometry clinic

Anti glaucoma agents

Antibiotics & anti-inflammatory drugs in ophthalmology

Radiology in ophthalmology

Diplopia

Ocular motor nerve palsy

Ptosis

CLINICAL SESSIONS

History taking, ophthalmic examination, slit lamp & hospital tour

Red eye, OSD (Ocular surface diseases) , common eyelid and lacrimal disorders

Cataract and glaucoma

Ocular motility, Strabismus & common pediatric eye diseases as leukocoria

Ophthalmoscopy, Uveitis and retinal diseases: diabetic retinopathy.

Refractive errors defective vision and amblyopia, Refractive surgeries

Eye manifestation of systemic diseases , thyroid eye diseases, Orbit

Emergency in ophthalmology ,endophthalmitis, Trauma and chemical burn.

ENT CASE STUDIES

CASE STUDY 1

A 23 year-old female, previously fit and well, complains of unilateral otalgia and an offensive discharge which has persisted for 2 days. She has recently been on holiday to Egypt.

1. What is the most likely diagnosis?
2. What are the most common organisms causing this condition?
3. She is planning on going on holiday again. What should you advise her about swimming and flying?
4. How would you expect the ear canal and tympanic membrane to appear?

5. What risk factors can predispose to this condition?

CASE STUDY 2

A six years old child developed severe pain in both ears together with a rise of temperature (39 C) following an attack of common cold. The child received medical treatment that lead to drop of his temperature and subsidence of pain; so the physician stopped the treatment.

1. What is the Diagnosis?

2. What is the most important examination step you have to perform?

However, the mother noticed that her child do not respond to her except when she raised her voice. This decreased response remained as such for the last 4 weeks after the occurrence of the primary condition.

3. Explain the following manifestations

-Ear condition following common cold

-Decreased response to sound

4. What are the investigations to be requested?

5. Mention the lines of treatment.

CASE STUDY 3

A three years old boy presented to the ENT specialist because of an inability to close the right eye and deviation of the angle of the mouth to the left side upon crying of 2 days duration. His mother reported that he had severe pain in the right ear 5 days prior to his present condition. She also added that his earache improved on antibiotic therapy.

1. What is the Diagnosis?

2. Explain the following manifestations

-Inability to close the right eye

-Deviation of the angle of the mouth to the left

-Onset of paralysis 5 days after the original condition

3. What is the most important examination step we have to perform?

4. What are the investigations should be requested?

5. Mention the lines of treatment.

CASE STUDY 4

A six year-old child presents with his father to the emergency ENT clinic with a pyrexia and complaining of otalgia. You note he has seen the school nurse recently as staff feel he is not working as hard as he was earlier in the year.

1. What diagnosis is most likely?
2. Why might he not be performing as well in school of late?
3. How would you expect the tympanic membrane to appear?
4. You examine the ear and notice that one ear appears to protrude more than the other. What complication are you worried about?

5. What common organisms are implicated in this condition?

6. List below the complications associated with infection in this part of the ear.

Intracranial Complications	Intra temporal bone Complications

He is started on intravenous antibiotics but after 24 hours of treatment there has been no improvement. After a normal MRI of his head to look for intracranial complications he is taken to theatre for a cortical mastoidectomy.

7. What important relations of the middle ear/mastoid are at risk?

PAGE 6
A thirty three

A thirty three year old lady presents with a one-year history of foul smelling discharge from her right ear and gradual deterioration in her hearing. On examination, there is a perforation at the attic, and a white substance occupies the area.

1. What is the most likely diagnosis?
2. What possible reasons are there for the deterioration in her hearing?
3. What results would you expect with tuning fork tests?
4. Post-operatively, you note that she is doing quite well, and she remains asymptomatic for the following two years. She returns some months later, with similar symptoms and new onset vertigo. What possible explanations are there for her latest presentation

CASE STUDY
PAGE 8

A 16-year old boy comes to the A&E department with epistaxis which has spontaneously resolved. He claims to have injured his nose whilst re-enacting last night's UFC match with a friend. He is seen by the A&E SHO and is sent home with no follow-up. He returns to the emergency ENT clinic 4 months later complaining of a change in shape to his nose. On examination, there is an obvious "step," which looks like a saddle deformity.

1. Describe the likely underlying series of events which have resulted in this deformity, after the initial trauma.
2. How would you have identified, and therefore prevented this from happening on his initial presentation?
3. What recreational drug is commonly implicated in nasal septal defects, and what is its method of damage?

CASE STUDY
PAGE 8

A 50-year old man presents to A&E with Epistaxis. He has tried conservative measures at home including ice, and pinching the soft part of the nose with no success. On arrival, he is tachycardic, tachypnoeic, and actively bleeding from his left nose

1. How would you manage this patient?
2. His nose is eventually packed which initially controls the bleeding. He then starts to spit up fresh blood. Outline what steps you would take to stop a posterior epistaxis.
3. How would you manage this patient if packing and cautery of the bleeding vessels were to fail?

CASE STUDY
PAGE 11

35 year old male patient presented with nasal obstruction with post nasal drip and frontal headache for 1 month duration ,the patient had similar attacks 3 months ago which respond to systemic and local treatment .Nasal endoscopy reveals pus in the left middle meatus with left side septal deviation

1-What is your diagnosis ?

2. .What are the lines of treatment ?

3. The patient do not respond to medical treatment and underwent FESS and septoplasty ?What investigation you have to send the patient for ?

4.. What are the drainage pathways of paranasal sinuses ??

CASE STUDY
PAGE 11

A 36-year old man with chronic rhinosinusitis is admitted for Functional Endoscopic Sinus Surgery (FESS).

1. What medical treatment would you have expected them to have had before considering surgery?
2. What are the potential complications of sinus surgery?

CASE STUDY
PAGE 11

30 year old male patients presented with bilateral nasal obstruction for the last 3 months ,his condition is temporarily relieved by vasoconstrictor nasal drops ,the patient had itching in the nose and throat with prolonged sneezing.

- 1- What other symptoms you should ask about for this patient?
- 2- Anterior rhinoscopy shows bilateral glistening masses with smooth surface with no bleeding ,no ulceration. What is your professional diagnosis ?
- 3- What are the lines of treatment ?
- 4- What are the adverse effects of vasoconstrictor nasal drops ?
- 5- How you will educate your patient about his illness ?

CASE STUDY 12

Sixty-five year old diabetic patient presented to the ENT outpatient clinic with left side nasal obstruction with attacks of mild epistaxis which was stopped by nose pinching ,On examination the left nasal cavity was filled with mass which has areas of necrosis and bleeding

1. What other symptoms you want to ask to reach the diagnosis ?
2. What other areas you want to examine and what you are looking for , Mention 3 areas ?
3. What investigation you will request and what are findings you expect to find ?
4. What other medical or surgical specialties can participate in the management of this patient ?
5. What are the lines of treatment ?

CASE STUDY
PAGE 16

Thirty-five year old female patient presented with dysphagia and high grade fever and inability to open her mouth , on examination the palatine tonsils were seen as red with deviation of the uvula to the left side .

1. What is your professional diagnosis ?
2. What other area you want to examine ?
3. The patient had right sided otalgia with normal ear examination. How you explain her ear symptom?
4. What are the lines of treatment ?
5. After one month she underwent tonsillectomy. What are the postoperative complications of this surgery ?

CASE STUDY
PAGE 16

Ten year old female patient presented with painful swelling of the left submandibular area for 5 days duration , on examination she was febrile looks in pain with huge tender fluctuant swelling of the left submandular area .

1. What is your diagnosis ?
2. What other area you want to examine and why ?
3. What are the possible complications of this condition ?
4. What are the lines of treatment ?

CASE STUDY
PAGE 16

Fifty one years old female was admitted to the hospital because of dysphagia of 2 months duration. The condition started by experiencing difficulty in swallowing of the solid food that arrested at the root of the neck. Then , and for the last few days, even liquids became difficult to be swallowed. She gave a history of throat lump sensation over the last 10 years. On examination, she had marked pallor of the mucous membrane of the oral cavity, glazed tongue and marked loss of weight. Examination of the neck revealed bilateral mobile hard upper deep cervical lymph nodes. The laryngeal crepitus is absent.

1. Explain the following manifestations;
 - a. Marked pallor of the mucous membrane of the oral cavity
 - b. Absent laryngeal crepitus
 - c. Bilateral mobile hard upper deep cervical lymph nodes.
2. From the disease progression , what is the type of this dysphagia?
3. What is the provisional diagnosis?
4. What is the procedure of choice to confirm the diagnosis?
5. If we decide to do surgery for this patient, what is the name of this surgical operation?

CASE STUDY
PAGE 16

Middle age heavy smoker presented to the otolaryngology clinic with a change in the quality of his voice for the last three months. The change was simple and intermittent , but , during the last two weeks the voice became constantly harsh and weak. The patient used to cough during that period but he used to experience this cough due to heavy smoking.

On examination , the patient looked dyspnic with audible respiratory sound , certain laryngeal examination was performed by the otolaryngologist when he decides to perform some other procedure but under general anaesthesia.

1. That do you call voice change?
2. What is the influence of smoking on the health of the respiratory system ?

3. Mention a clinical examination to assess the gross appearance and function of the larynx ?
4. If you decide to get a biopsy from the larynx , how can you perform that " the procedure "?
5. Suppose that the pathology became more severe and compromise the airway , what steps you have to follow to save the patient's life ?

OPHTHALMOLOGY CASE STUDIES

CASE STUDY 1

A 70 year old man comes in complaining of painless loss of vision in his right eye. His past medical history includes type 2 diabetes (diet controlled) and hypertension. He has had one myocardial infarction three years ago.

On examination you find that he has a right relative afferent pupillary defect. Name 3 conditions in which you would see a RAPD.

On Ophthalmoscopy you see the following:



There are widespread retinal haemorrhages and a few soft exudates in the right fundus. The Left eye is normal.

<http://emedicine.medscape.com/article/1223746-overview>

A- What is the diagnosis?

B- What are the possible outcomes for this patient without treatment?

CASE STUDY 2

A 5 year old boy presented by his parents because of ocular misalignment since the last 6 months where they notice constant ocular deviation

A. the parents are worry about their child ,are they right?

B. why the constant deviation is more important than intermittent.

C. if visual acuity and refractive examination were done ,the child was hyperopic what is the possible treatment.

D. the visual acuity was decreased, what is the expected causes?

E. ocular motility was done and looked normal, what is the importance of this test?

F. if the surgeon decide to do surgery to correct the deviation what ocular muscle should be weakened ? suppose the child has convergent squint.

CASE STUDY 3

A 2 year old baby presented with drooping of right upper lid(ptosis) since birth, inspite of uncooperative child the surgeon could examine eyes and were normal , red reflex can be .seen clearly the patents wanted surgery but the surgeon decided to postponed the surgery

A. what are the muscles which responsible for lid elevation?

B. what is the importance of red reflex in this case?

C. why the surgeon postponed the surgery?

D. what are the determinants of type of surgery?

E. indicate the causes of pseudo-ptosis.

CASE STUDY 4

A 2 year old baby presented because of white pupil (leukocoria) in right eye , with no history .of pain or redness and no family history

A. give five differential diagnosis.

B. what is your important steps for the provisional diagnosis?

C. for prevention of the above cause, what effort should be done?

D. if the parents refuse the treatment , what you have to tell the parents about the future?

E. describe the types of treatment.

F. why the pupil become white, what is the original color physiology.

CASE STUDY

PAGE 6

A 27 year-old lady comes into the emergency department complaining of severe eye pain, watering and sensitivity to light, having splashed oven cleaner in her eye.

- A. What is your immediate management?

- B. Which type of chemical injuries are more severe and why?

- C. On examination you see the following:
 - o Red eye, expect for pallor of vessels close to the limbus in the interpalpebral fissure
 - o Cloudy cornea
 - o Hazy view of Iris detail

- D. What is the cause of the unusual conjunctival injection?

- E. What further treatment is used to treat this condition?

- F. What are the long-term complications?

- G. Does this patient need urgent referral to eye casualty? Please explain your answer

CASE STUDY
PAGE 6

A 70 year-old lady presented in A&E with a one day history of sudden, painful loss of vision, headaches and nausea developing over the course of 24 hours.

- A. Name a likely diagnosis?
- B. You find out that she has a history of being long sighted, needing reading glasses first at the age of 55 years. On examination what **classic signs** would you look for:
- C. What medical treatment would need to be immediately initiated?
- D. Give two systemic medications or diagnostics drugs that could induce this condition

CASE STUDY

PAGE 10

A 45 year-old man who is asymptomatic, is referred to you by his optician with probable chronic simple glaucoma

- A. What is the definition of Glaucoma?
- B. What risk factors would you ask about in your history?
- C. What are the characteristics of the visual field loss:
- D. On examination what other features would support your diagnosis?
- E. What classes of medication would you use to treat Mr Jones & give an example of each.
- F. What surgical treatment is available if medical therapy is ineffective?
- G. What are the principles of Glaucoma surgery?

CASE STUDY

PAGE 10

year girl notice a progressive change in her vision with frequent change in her glasses 16 years old. In the last 6 months, the ophthalmologist suspects that the patient might have keratoconus.

- A. Why does the doctor suspect keratoconus? Explain the pathophysiology?
- B. Mention three factors which are responsible for corneal integrity.
- C. The doctor asks the patient to do corneal topography and pachymetry, why?
- D. If the diagnosis proved to be keratoconus, how do you stop this condition?
- E. If you are sure that the condition will not become progressive, what are the principles of visual correction?
- F. What is a corneal graft?

PAGE 10
A 47-year-old woman

A. What is your diagnosis?

B. On examination, her eyelid margins are red and inflamed and on fluorescein staining you find that she has punctate staining on the inferior third of the cornea.

C. What is your initial advice about managing the condition?

D. What further medical treatment may be appropriate?

CASE STUDY

PAGE 10

Helen, a 4 year girl is referred to clinic for an eye check. On slit lamp examination you see flare and cells in the anterior chamber.

A. What does the picture show?

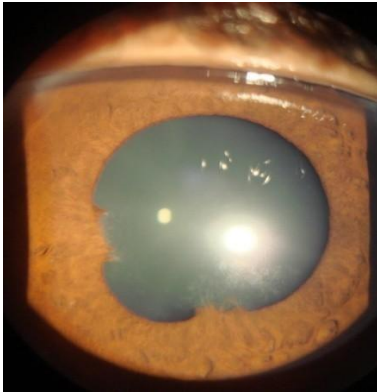


Image taken from Visoncarecentre.com

B. What is your Ophthalmological diagnosis?

C. What underlying disease do you suspect?

D. What treatment is required for her eyes?

E. What long-term ophthalmological complications may develop?

F. Record a history from a patient with a history of eye discomfort or visual disturbance that has developed within the previous 2 weeks. Formulate a provisional diagnosis and examine the patient to test your diagnostic hypotheses. Describe briefly the appropriate management:

G. Take a history from a patient with a history of eye discomfort or visual disturbance that has been present for more than 2 weeks. Formulate a provisional diagnosis and examine the patient to test your diagnostic hypotheses. Describe briefly the appropriate management:

H. Ask a patient with significant long term visual or hearing impairment about the impact this has on their daily life. Describe how these difficulties are addressed.

CASE STUDY 11

A 71 year old man is referred by his optometrist with reduced visual acuity associated with a substantial change in his spectacle prescription over the last six months. He tells you he is having problems driving at night and often gets glare when looking at oncoming vehicles.

- A. What is the most likely causes for his symptoms?
- B. How can his change in prescription be explained?
- C. You decide that he is a suitable candidate for surgery, what operations are available for his eye condition?
- D. At roughly what level of vision do most people undergo surgery?
- E. The patient asks you what the operation involves, write down here what you would tell him.
- F. What complications of sugery would you warn him of?
- G. What percentage of world blindness does this condition account for – according to WHO?
- H. List the other commonest causes of blindness worldwide and the treatment & public health measures that can be taken to treat or prevent them

CASE STUDY 12

A 75 year old lady presents complaining of difficulty reading for the past two months. On further history taking she explains that she finds it hard to see her sons face clearly. What two investigations can be used to investigate the diagnosis?

Fill in the table below with the following information: What are the two subtypes of this disease How can you differentiate between them What treatment is available

		1)	2)

A barber working 12 hours daily standing, presented with bilateral erythematous dry

Learning opportunities

During this block students will have opportunities to acquire knowledge in the following either through self-study, lectures or clinical observation:

Basic sciences:

- Anatomy of the special senses *Pharmacology & therapeutics*
- Review of sympathetic and parasympathetic nervous system
- Pharmacology of nasal decongestants.
- Treatment of acute eye infections
- Treatment of acute ear infections

Infection

- Acute otitis media
- Sinusitis
- Conjunctivitis
- Corneal Abscess

Imaging

- Radiology of sinuses

Assessment of the block

At the end of the block a report will be made to the Medical School as to whether each student has:

- Attended all compulsory timetabled sessions; and other clinical sessions as appropriate
- Completed the required tasks defined in the block workbook
- Satisfactorily completed in-block formative assessments (as appropriate for the block), including consultation skills and knowledge-based assessments.
- Demonstrated professional attitudes and behaviour

ENT

A. The Ear

Learning Objectives:

- To revise the gross anatomy of the ear, and the relevance of this to clinical practice.
- To understand the pathophysiology, and therefore management of common conditions involving:
 - o The external ear
 - o The middle ear
 - o The inner ear



The anatomy of the ear is complex, especially when considered as a single entity. It is much more palatable when considered in three parts; external, middle and inner ear.

This chapter aims to cover the common clinical conditions in each of the three parts of the ear, thus supplementing the anatomy and physiology of the head and neck as taught in the Applied Head & Neck module in Phase 1.

1. External Ear

a) Ear Trauma

This is a common occurrence, typically seen in sport-related injuries, but also as a result of violence and aggression involving a blow to the ear. Fortunately in the majority of uncomplicated cases, treatment can be administered quickly and often under local anaesthetic.

It is therefore important to know the sensory supply to the pinna.

- Upper lateral surface – CN V₃ – Auriculotemporal nerve.
- Lower lateral surface and medial surface – C3 – Greater auricular nerve.
- Superior medial surface – C2/C3 – Lesser occipital nerve.
- Auricular branch of vagus – External Auditory Meatus

Performing regional nerve blocks at each of the nerves mentioned above, anaesthetises the pinna and allows a variety of procedures to be performed.

Lacerations

Simple primary closure of the skin with sutures after adequate cleaning is all that is required in most cases. It is important to ensure that any exposed cartilage is covered with skin. If there is significant skin loss where primary closure will not be possible an opinion from a plastic reconstructive surgeon may be required.

(ii) Bites

Bites to the ear carry a significant risk of infection from skin commensals, or oral commensals from the offending creature/person. These require an appropriate history to ascertain likely organisms involved in potential infection, and the wound must be left open. Treatment includes thorough wound irrigation and antibiotics.

(iii) Haematoma

Pinna haematomas cause disruption in the blood supply to the cartilage. Normally, the cartilage obtains nutrients via diffusion from vessels in the overlying perichondrium, and a disruption can therefore lead to a vascular necrosis.

There is a significant risk of associated deformity (cauliflower ear) and this therefore requires urgent drainage and pressure dressing application to prevent re-accumulation.



(iv) Tympanic Membrane Perforation

The tympanic membrane can be perforated by direct or indirect trauma. This subsequently leads to:

- Pain
- Possible conductive deafness

Most perforations will heal by themselves, and therefore only require a conservative “watch and wait,” approach with the patients following water precautions. In cases where the perforation does not heal after a period of 6 months or more the patient may benefit from surgical intervention in the form of a myringoplasty to repair the tympanic membrane if the perforation is causing problems.

Perforations can also be caused by otitis media.

(v) Haemotympanum

Trauma may cause a haemotympanum (blood in the middle ear) which may be associated with a temporal bone fracture. This can be seen through the tympanic membrane and is associated with a conductive hearing loss. It should be treated conservatively as it will settle with time but patients should be followed up to ensure that there is no residual hearing loss from damage to the ossicles.

B) Ear Infections (Otitis Externa)

This is inflammation of the skin lining the external canal, and is occasionally known as “swimmer’s ear”. It may be bacterial or fungal. When considering otitis externa (OE), it is useful to consider the external ear as a continuation of the rest of the skin. It is therefore inhabited by the same skin commensals that can be found elsewhere.

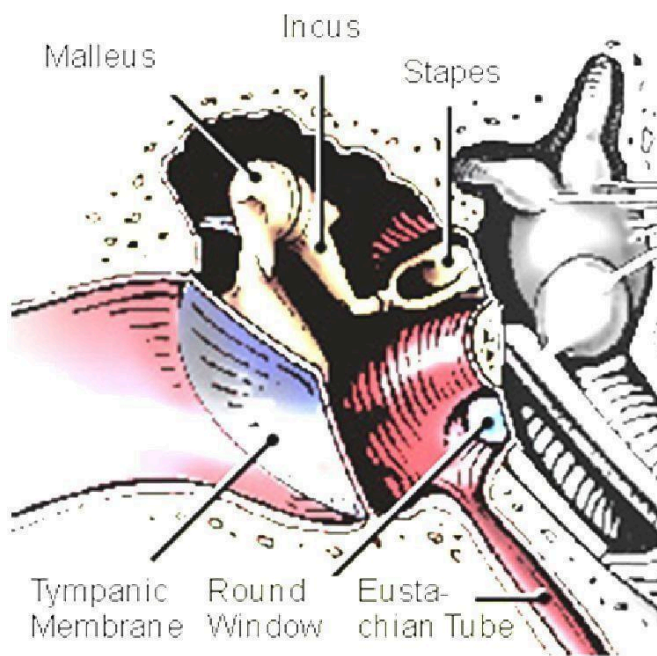
Patients normally present with a painful discharging ear and there is often a history of an itchy ear. The hearing may be muffled from the discharge present in the ear canal.

Malignant otitis externa is a particularly aggressive infection mainly seen in diabetics or immunocompromised patients where the infection spreads from the soft tissue of the ear canal into the bone. It normally presents with chronic ear discharge despite topical treatment, a deep seated severe ear pain and sometimes cranial nerve palsies (most commonly CNVII). It has a significant mortality rate of around 10% despite aggressive treatment.

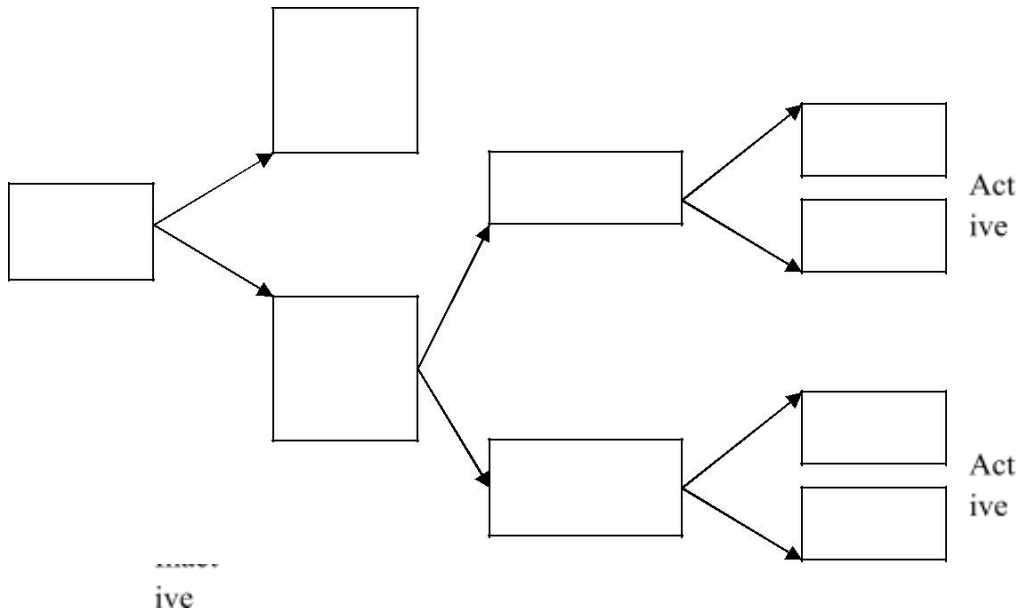
Management

- Topical eardrops empirically e.g. Gentamicin.
- Swab any discharge in resistant cases.
- Microsuction of pus/debris which enables the drops to get to the source of the infection
- In severe infection, a wick may be used to help hold the canal open and to allow topical treatment to diffuse through
- Fungal infections should be treated with topical antifungals as appropriate.
- Malignant otitis externa will require aggressive treatment with iv antibiotics as well as topical treatment for an extended period of time to eradicate infection

2. Middle Ear



a) Ear infections (Otitis media)



(i) Acute Otitis Media

Acute otitis media (AOM) is an infection of the middle ear. Unlike the external ear, the epithelium lining the middle ear is respiratory epithelium, or pseudostratified columnar epithelium. The middle ear can therefore be regarded as a continuation of the upper respiratory tract, and is therefore susceptible to a similar variety of pathogens.

AOM is more common in childhood and is related to eustachian tube dysfunction. Common pathogens include *Streptococcus pneumoniae*, *Haemophilus influenzae* and *Moraxella* species.

Clinical Features

- Ear pain
 - caused by increased pressure in the tympanic cavity, in young children this may be evident by ear pulling.
- Discharge
 - the tympanic membrane may rupture with the pus from the middle ear discharging into the ear canal. When this happens the pain settles and the ear discharge starts.
- Fev

er Management

- Conservative – Most patients can be managed conservatively with analgesia

- Medical – In severe or persistent cases oral antibiotics may be required
- Surgery – Recurrent AOM may be helped by grommet (ventilation tube) insertion

(ii) Chronic Otitis Media

Chronic otitis media (COM) can be divided into active or inactive depending on whether the ear is discharging (active) or not (inactive). It can then be subdivided into mucosal disease or squamous disease. Active squamous disease is also known as cholesteatoma and inactive squamous COM is a retraction pocket which may develop into active disease (cholesteatoma). Active mucosal disease is where there is chronic discharge from the middle ear through a tympanic membrane perforation and inactive mucosal disease is where there is a tympanic membrane perforation but no active infection/discharge.

Mucosal COM is thought to develop from an episode of AOM where after rupturing of the tympanic membrane there is a failure to heal and active squamous COM is thought to develop when keratinised squamous cells are introduced into the middle ear from a retraction pocket or a perforation.

Active COM is associated with chronic ear discharge and often a conductive hearing loss. It can also be associated with complications associated with spread of the disease within the temporal bone and intracranially.

Management

Management of COM depends on whether there is cholesteatoma present (squamous disease) or not. If there is cholesteatoma present surgery is required to clear the cholesteatoma. The cholesteatoma will often grow from the middle ear into the mastoid bone and this therefore often requires surgery to the mastoid bone (mastoidectomy). If there is no cholesteatoma (mucosal disease) there is a chance that this may clear up with medical treatment (topical antibiotic drops and aural toilet). Often it is not known whether cholesteatoma is present in a chronically discharging ear and in this case the ear should be treated medically first and then if this fails surgery should be carried out when cholesteatoma may or may not be discovered. In surgery where there is no cholesteatoma found the aim is to repair the perforation and ensure good ventilation of the middle ear and mastoid bone but it is not necessary to remove every last trace of disease unlike cholesteatoma. Mastoid surgery is associated with a small risk of a facial nerve palsy, altered taste from damage to the chorda tympani, CSF leak, tinnitus, vertigo and also of complete loss of hearing in the operated ear.

b) Otitis media with effusion (Glue ear)

This is a condition where fluid is present in the middle ear with an intact tympanic membrane and is related to eustachian tube dysfunction. It is more common in children but can occur in adults. In adults with an Otitis Media with Effusion (OME) (especially unilateral) it is very important to have a look at the post nasal space as tumours in this area can cause eustachian tube dysfunction leading to OME. OME is not painful but the middle ear may become infected which will lead to an acute otitis media which is painful.

Clinical features

- Middle ear effusion on otoscopy
- Conductive hearing loss
 - o This may be associated with speech delay and problems at school

Investigations

- Tympanogram
 - o Flat (Type B) Tracing with normal canal volume
- Pure tone audiogram
 - o Conductive hearing loss

Management

- Conservative – most cases settle within 3 months
- Hearing aid
- Surgery - for prolonged hearing loss causing significant problems o Grommets (ventilation tubes)
 - o +/- Adenoidectomy

c) **Otosclerosis**

This is a disease affecting the ossicles in the middle ear. The cause has both genetic and environmental components. In families where otosclerosis is prevalent it appears to follow an autosomal dominant transmission. Mature bone is gradually replaced with woven bone, and symptoms develop as the stapes footplate becomes fixed to the oval window.

Otosclerosis causing symptoms is thought to occur in 1-2% of the population and 85% of these will develop bilateral disease. Twice as many females are affected compared with males.

History

- Progressive hearing loss
- Often tinnitus
- Improved hearing in noisy surroundings during early stages of disease
- Family History

Examination

- Most commonly normal
 - Rarely pink hue to the tympanic membrane – Schwartz's sign

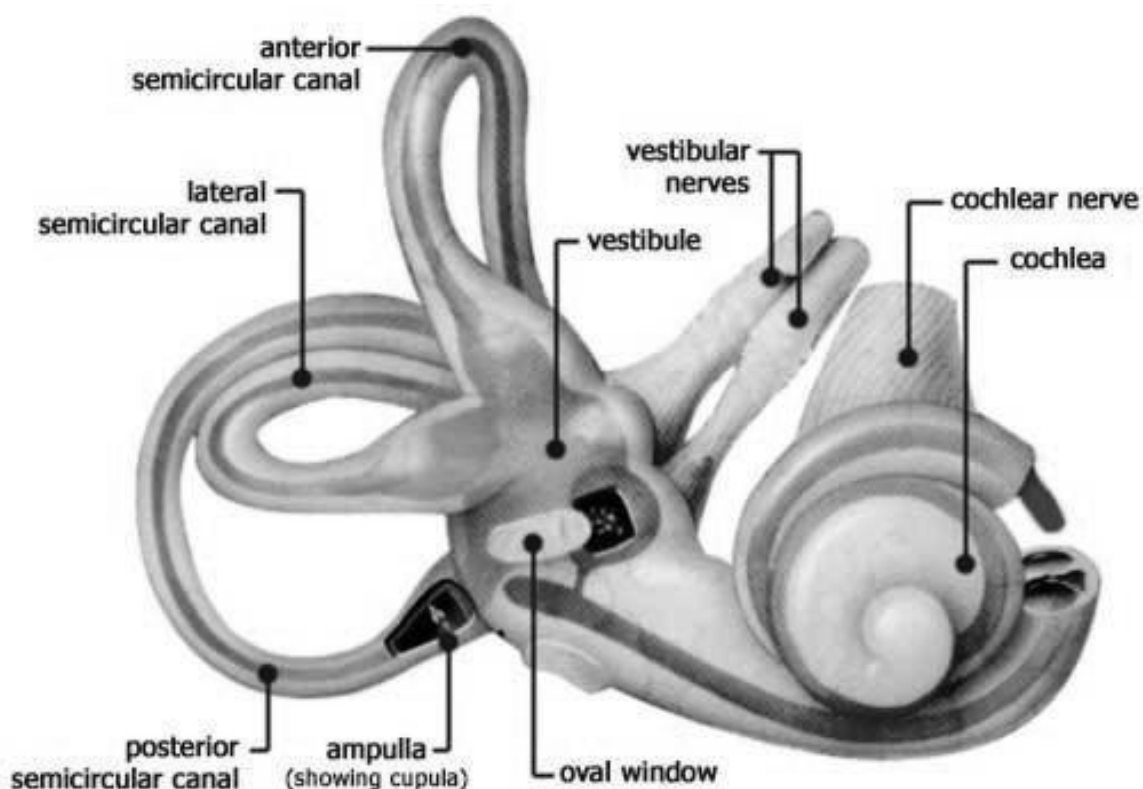
Investigations

- Tympanogram
 - Normal type A trace
- Pure tone audiogram
 - Conductive hearing loss
 - A characteristic "Carhart notch" at 2kHz

Management

- Conservative – hearing aid
- Surgery - stapedectomy

3. Inner Ear



The membranous labyrinth is filled with endolymph, and this in turn is surrounded by the bony labyrinth. The membranous labyrinth is suspended in perilymph, contained within the bony labyrinth.

The perilymphatic system communicates with the subarachnoid space and CSF via the cochlear aqueduct. Perilymph fluid resembles CSF, and endolymph fluid resembles intracellular fluid.

The cochlea is responsible for the perception of hearing and is analogous in structure to the rest of the labyrinthine system but has a number of distinguishing features and functions;

- It has 2.5 turns around a bony core, the modiolus.
- The stapes articulates with the oval window, causing movement of perilymph, and a pressure change, compensated by the round window.
- Vibrations are transmitted through the endolymph to the tectorial membrane.
- Low frequency sounds are detected at the apex of the cochlea, and high frequency sounds at the base of the cochlear.
- Movement of the tectorial membrane causes movement of hair cells, and subsequent depolarisation of neuronal fibres allowing perception of sound.
- Transmission of information occurs via the cochlear nerve.

The vestibular system can be further divided into:

- Semicircular canals
- Utricle
- Saccule

There are three semicircular canals, each existing at 90° to the other.

The utricle and saccule detect movement in a linear fashion, rather than the rotatory mechanism as seen in the semicircular canals.

Simply put:

- **U**tricle – Hair cells point **U**p – Detect linear/horizontal movement
- **S**accule – Hair cells stick out to the **S**ide – Detect vertical movement

Good balance requires input from the vestibular system to be integrated centrally with proprioceptive and visual inputs and problems in any of these areas can cause problems with balance.

a) Vertigo

Vertigo is defined as the hallucination of movement and is associated with a problem with the vestibular system. It is essential to take a thorough history, and hence ascertain potential causes and treatment options. Causes may be due to central or peripheral vestibular pathology;

- Central causes:
 - o Stroke
 - o Migraine
 - o Neoplasms
 - o Demyelination eg. MS
 - o Drugs
- Peripheral causes:
 - o BPPV
 - o Ménière's disease
 - o Vestibular Neuronitis

(ii) BPPV

Benign Paroxysmal Positional Vertigo, as its name suggests, is vertigo occurring with particular head movements, which is benign in nature, and lasts a short amount of time, typically seconds. Unfortunately, this is often a very distressing condition for the patient, and at times, disabling.

The underlying cause are otoliths (crystals) in the semicircular canals (most commonly posterior) causing abnormal stimulation of the hair cells giving a hallucination of movement i.e. vertigo.

Diagnosis is by the Dix-Hallpike test and treatment is with the Epley manoeuvre. You should have the chance to see both of these during your attachment.

(ii) Ménière's Disease

The precise aetiology of this is still unknown although many theories exist. The pathophysiology involves endolymphatic hydrops (increased fluid in the endolymphatic compartment).

Clinical Features

- Tinnitus in affected ear
- Episodic vertigo lasting minutes to hours associated with nausea & vomiting
- Fluctuating sensorineural hearing loss which over time becomes permanent
- Aural fullness

Initially in the course of the disease patients are well between attacks but as the disease progresses they may feel generally unsteady between attacks as they start to develop reduced vestibular function on the side of the disease and a progressive sensorineural hearing loss. Over time the disease often burns itself out so that the patient doesn't suffer from acute vertigo but will have reduced hearing and maybe generally unbalanced. If the vestibular system of the other ear is working well this can compensate for the bad ear to improve the overall balance but this can take time especially in the elderly.

Management

- Dietary – reduce salt, chocolate, alcohol, caffeine, chinese food
- Medical
 - Thiazide diuretics e.g. bendrofluazide
 - Betahistine
 - Vestibular sedatives e.g. prochlorperazine for acute attacks only
- Surgical
 - Grommet insertion
 - Dexamethasone middle ear injection
 - Endolymphatic sac decompression
 - Vestibular destruction using middle ear injection of gentamicin
 - Surgical labyrinthectomy – very rarely required

(iii) Vestibular Neuronitis

In this condition inflammation in the inner ear causes severe incapacitating vertigo lasting several days associated with nausea and vomiting. During an attack patients will have horizontal nystagmus but an otherwise normal neurological examination. Treatment is symptomatic with vestibular sedatives during the acute episode and intravenous fluids if required. There is often an associated long term vestibular deficit after the acute episode which can lead to a generalised unsteadiness after the acute episode for a number of weeks while the brain compensates for this. If patients are suffering from prolonged poor balance from vestibular hypofunction in one ear from vestibular neuronitis or any other pathology they can be helped by vestibular rehabilitation exercises such as the Cawthorne-Cooksey exercises. After the acute attack patients should not take vestibular suppressants as this will delay their recovery.

b) Sudden onset sensorineural hearing loss

This is an otological emergency that may cause lifelong disability for the patient. Patients who have a sudden onset hearing loss need to be assessed to see whether it is sensorineural or conductive. This can be done by tuning fork tests or a pure tone audiogram if you have access to this.

Prognosis

1. 1/3 recovery to normal
2. 1/3 some recovery
3. 1/3 no

recovery

Investigations

- Pure tone audiogram
- MRI scan
 - o to exclude lesion along central auditory pathway. e.g. acoustic neuroma

Management

Steroids – normally orally but can be injected into middle ear Anti-virals

Other treatments are used e.g. hyperbaric oxygen, carbogen but are not widely practiced

4. Hearing Tests

a) Tuning fork tests

These tests help to differentiate a sensorineural hearing loss from a conductive hearing loss. This is extremely important as patients with sudden onset sensorineural hearing loss require urgent treatment and most of the time that these patients present you will not have access to a pure tone audiogram. A 256 Hz or 512 Hz tuning fork should be used. These tests assume that there is normal hearing in the other ear.

(i) Weber Test

The tuning fork is knocked against your knee or elbow and placed on the patient's forehead. The patient is asked if they can hear the noise on the left, right, centre or not at all. The tone produced from the tuning fork is transmitted via the bone of the skull to both cochlear.

- Sensorineural hearing loss – the tone will be heard on the opposite side to the hearing loss
- Conductive hearing loss – the tone will be heard on the same side as the hearing loss. This is because the conductive loss will block out background noise so relative to the other ear the tone will sound louder in that ear

(ii) Rinne test

The tuning fork is knocked against your knee or elbow and placed on the patient's mastoid for a few seconds and then lateral to the external meatus of the ear. By placing the tuning fork on the mastoid the tone is being conducted to the cochlear via the temporal bone and by placing the fork lateral to the ear the tone goes through the external canal and gets the benefit of the amplification from the external and middle ear. In a normal hearing patient the tuning fork will be heard louder when held lateral to the external auditory meatus, this is referred to as Rinne positive as opposed to Rinne negative where the sound is heard louder when the tuning fork is placed over the mastoid.

- Sensorineural hearing loss – Rinne +ve if heard at all because you are still getting the benefit of the amplification of the external and middle ear
- Conductive hearing loss – Rinne -ve as you are no longer getting the benefit of the amplification from the middle and external ear

b) Pure tone audiogram (PTA)

This test assesses the hearing thresholds of patients at different frequencies and is the most widely used assessment of patient's hearing. It can be used from around the age of 4 upwards. It should be performed in a sound proofed booth. Tones are played to the patient at different frequencies to each ear. Each frequency is tested a number of times and the quietest tone that can be reliably heard is marked on the audiogram.

The frequency in hertz is on the x axis and the decibel scale on the y axis with 0 at the top and increasing decibels (loudness) as you descend the y axis. The higher the decibel number the louder the noise, so the higher the line on the pure tone

audiogram the better the hearing. Anything above 20dB on the audiogram is considered normal. Air conduction and bone conduction thresholds are tested.

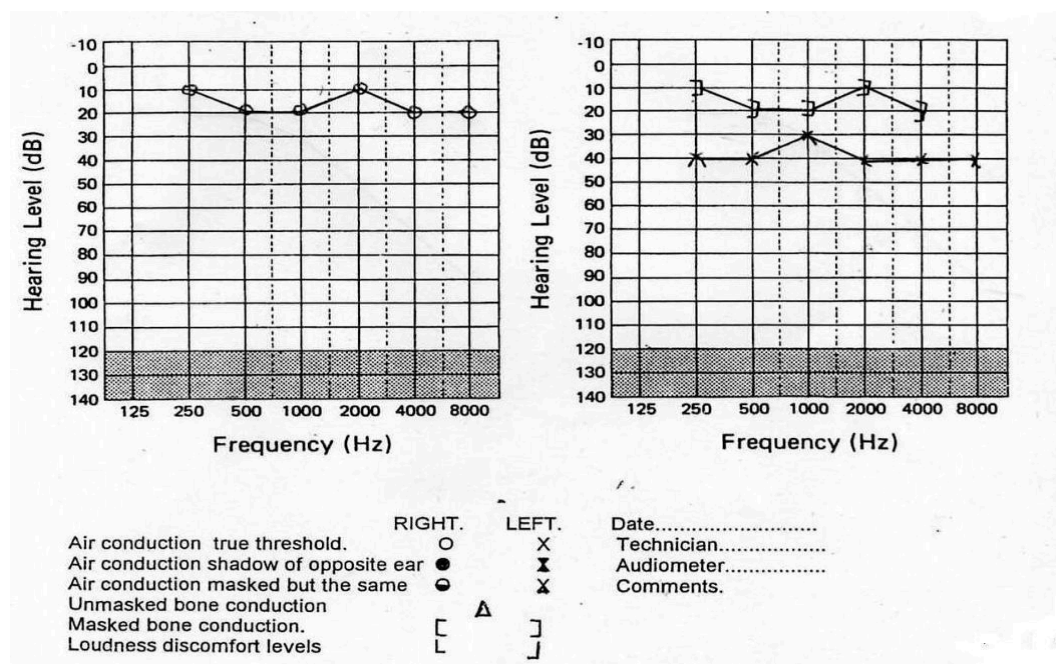
Air conduction is measured by playing the tone through headphones and assesses the whole auditory pathway. Bone conduction is measured by playing the tone through a bone conductor placed over the mastoid bone. This then passes through the bone straight to the cochlear bypassing the conductive part of the auditory system (external and middle ear) and therefore approximates to sensorineural hearing. Sound played through the bone conductor to one ear will not only travel to the cochlear on that side but also to the cochlear on the other side so if there is any discrepancy in the hearing between the ears the other side will need to be masked. This means playing a noise to the contralateral ear to prevent it from detecting the test tone. This is not such a problem for the air conduction unless there is a very large difference in air conduction thresholds between the two ears.

The thresholds are marked on the audiogram. Different marks are made on the audiogram depending on the ear being tested, whether bone conduction or air conduction is being recorded and whether masking is being used or not. This information will normally be on a legend beside the audiogram. Both ears may be marked on the same graph or they may be plotted on separate graphs next to each other when the right ear will be on the left hand graph and the left ear on the right hand graph.

There are two types of hearing loss – conductive and sensorineural.

(i) Conductive

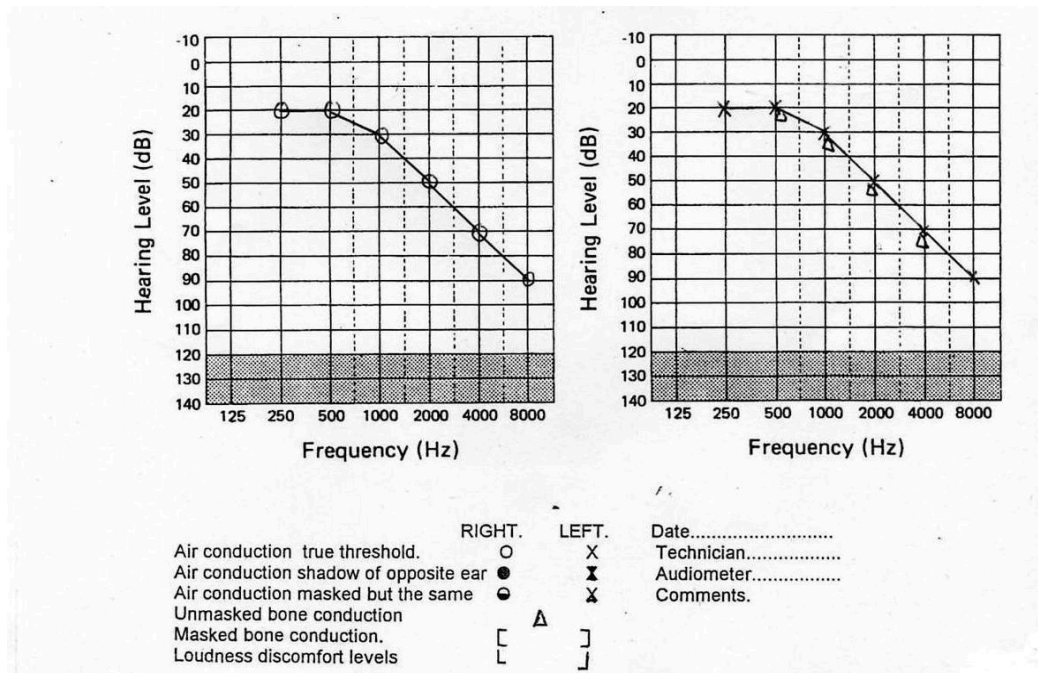
- this is caused by an impedance to hearing in the middle or external ear
- e.g. wax, otitis media with effusion
- the audiogram has normal bone conduction and reduced air conduction thresholds i.e. an air bone gap



Pure tone audiogram showing normal hearing in the right ear and a conductive hearing loss in the left ear.

(ii) Sensorineural

- this is caused by a problem from anywhere between the cochlear and the auditory cortex of the brain
- asymmetrical sensorineural hearing loss should be investigated with an MRI scan looking for lesions along the pathway e.g. acoustic neuroma (vestibular schwannoma)
- the audiogram will have reduced bone and air conduction thresholds i.e. no air bone gap



Audiogram showing a bilateral sensorineural hearing loss consistent with hearing loss seen in old age (presbycusis)

Patients may also have a mixed hearing loss with both conductive and sensorineural components.

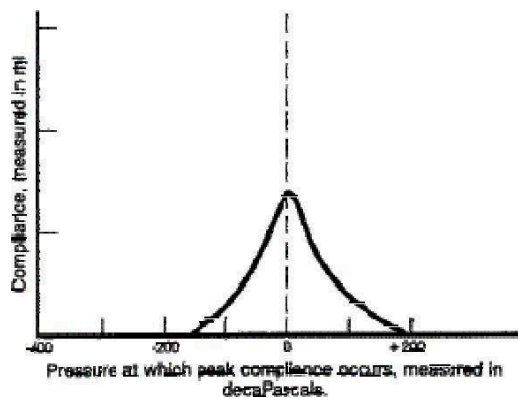
c) Tympanogram

This is a simple test that measures compliance of the tympanic membrane and gives useful information about the tympanic membrane, the middle ear and eustachian tube function. It involves inserting a probe in the external ear canal and only takes seconds to perform. It can be done at any age as it takes minimal cooperation from the patient.

Compliance is measured along the y axis and pressure (daPascals) is measured on the x axis. The compliance of the tympanic membrane is then measured with varying amounts of pressure in the external ear canal which is sealed by the probe. The compliance peaks when the pressure in the canal equals that of the middle ear. Three types of tracing are classically seen

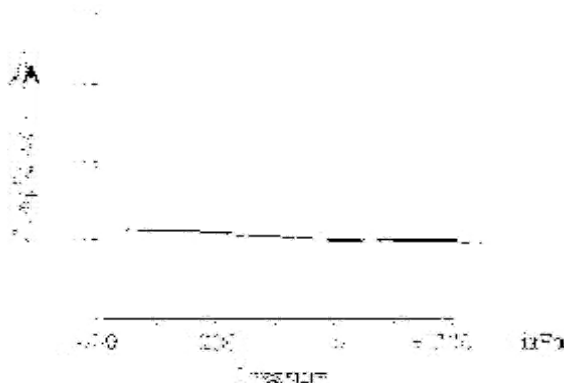
(i) Type A

- Normal result
- Peak centred on 0 daPa on x axis



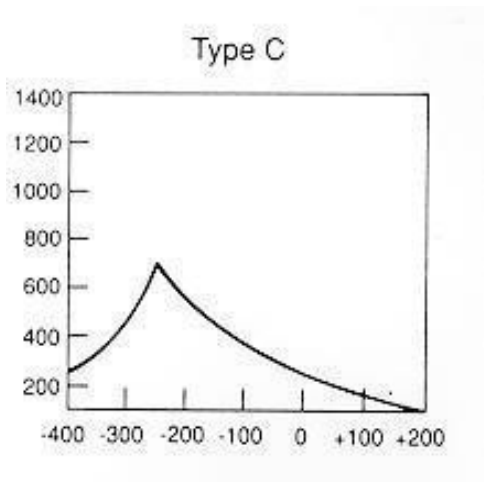
(ii) Type B

- Flat tracing
- Suggests middle ear effusion or perforation
- By looking at the canal volume reading on the side of the tympanogram you can differentiate an effusion from a perforation. An effusion will have a normal canal volume (around 1 cm³ in adults) whereas in a perforation the volume being measured will be much larger as it is measuring the middle and external ear volume.



(iii) Type C

- Suggests eustachian tube dysfunction
- The peak of the tracing has negative pressure



B. The Nose/Sinuses

Learning Objectives:

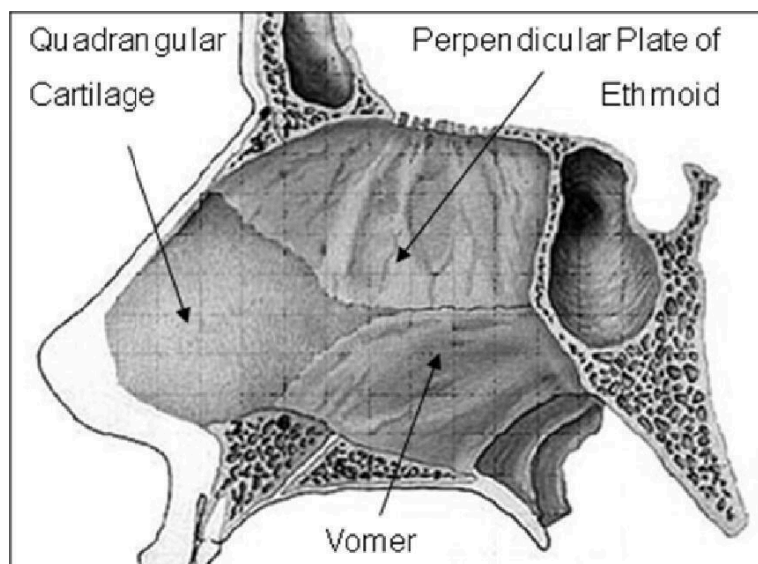
- To revise the gross anatomy of the nose and paranasal sinuses, and their clinical relevance.
- To understand the pathophysiology of common conditions affecting the nose and paranasal sinuses.

This chapter aims to cover the common clinical conditions you will see affecting the nose and related structures. Anatomy will briefly be revisited. The nose will be considered as one entity, with the paranasal sinuses being discussed separately, and a brief mention of the nasopharynx and its clinical relevance.

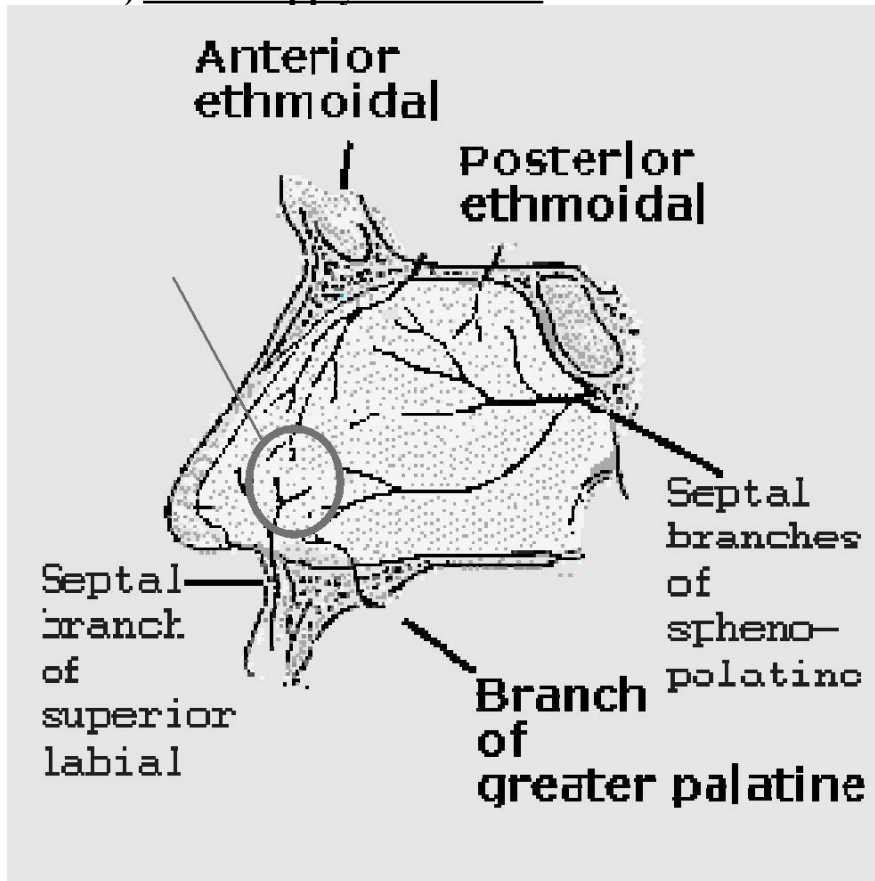
1. Anatomy of the Nose

a) External nose

The external nose receives support from bone (nasal bones) and a number of cartilages.



b) Blood Supply to the Nose



Nasal Septum

Little's area, also known as Kiesselbach's plexus, is a highly vascular area on the nasal septum. It receives a contribution from branches of the external carotid and internal carotid arteries.

This becomes important when considering the management of epistaxis, as haemorrhage may be from one, or a combination of these arteries.

2. Epistaxis

Epistaxis is bleeding from the nose, and in the vast majority of cases is relatively insignificant. In a small proportion however, it can present with massive haemorrhage and shock.

It is vital therefore to obtain a concise, but thorough history to establish the cause, and any perpetuating factors such as anticoagulant use, and hypertension.



Local

- Idiopathic – 85%
- Traumatic
- Iatrogenic
- Foreign Body
- Inflammatory – Rhinitis, Polyps
 - Neopl

astic

Systemic

- Hypertension
- Coagulopathies
- Vasculopathies
- Hereditary Haemorrhagic Telangiectasia/Osler-Weber-Rendu disease

Management

- ABC
- First Aid
 - o Pinch soft part of nose
 - o Head forward
 - o Spit out (not swallow) any blood in mouth
- Examination
 - o Locate source of bleed – Anterior or Posterior bleeding?
- Conservative
 - o Cautery – silver nitrate or bipolar diathermy
 - If anterior bleed – with anterior rhinoscopy
 - If posterior bleed – with rigid endoscope
 - Topical adrenaline may help control bleeding before cautery
 - o Nasal packing if cautery fails to control bleeding
 - Initially anterior pack
 - If continues bleeding into oropharynx anterior & posterior pack
- Surgical/Radiological
 - o If nasal packing fails to stop the bleeding the following vessels can either be ligated surgically or embolised radiologically
 - Sphenopalatine
 - Anterior ethmoid (can not be embolised because comes from internal carotid artery)
 - External carotid (last resort)

3. Nasal Trauma

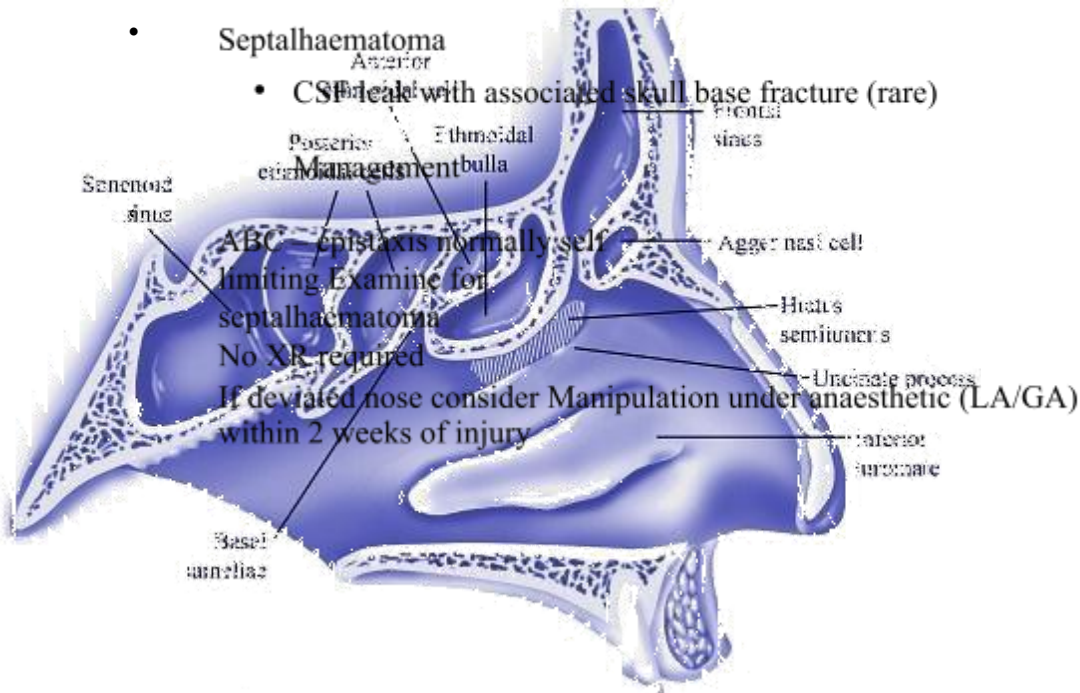
Nasal trauma commonly occurs following assault, and is also seen in sport, falls, and



road traffic accidents.

Fracture of the nasal bones may be complicated in the following ways:

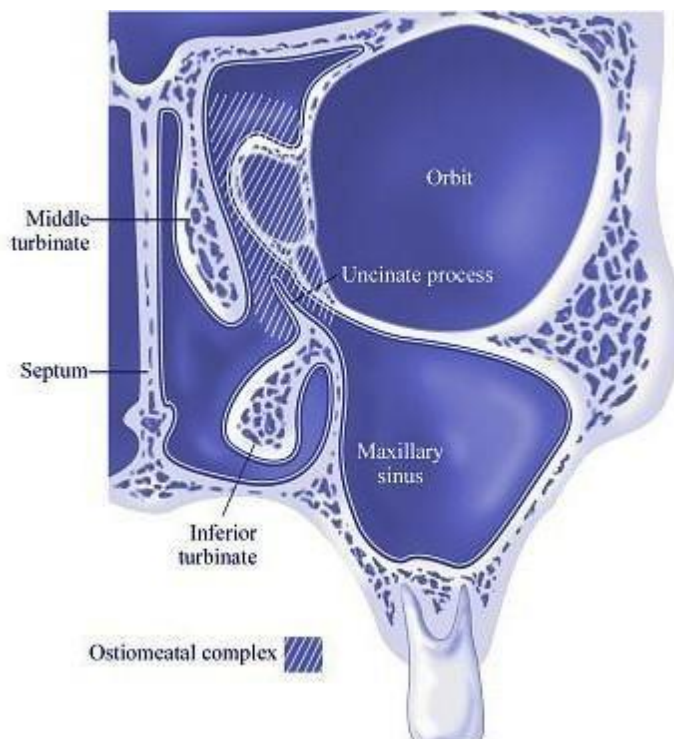
- Septal haematoma
 - CSF leak with associated skull base fracture (rare)
- Management**
- ABC – epistaxis normally self-limiting. Examine for septal haematoma. No XR required. If deviated nose consider Manipulation under anaesthetic (LA/GA) within 2 weeks of injury.



4. Paranasal Sinus Anatomy

Knowing the relationship of the paranasal sinuses is particularly important when considering surgery.

Due to the intimate nature of structures within the head, there is a significant risk of injury to surrounding structures, especially when the surgeon is working with minimal access in such an enclosed space.



Structures of importance include the lamina papyracea forming the medial wall of the orbit, the anterior cranial fossa and the internal carotid artery.

Site of Drainage	Paranasal Sinus
Spheno-ethmoidal recess	Sphenoid sinus
Superior meatus	Posterior ethmoid cells
Middle meatus	All other sinuses • Frontal • Maxillary • Anterior ethmoid cells
Inferior meatus	Nasolacrimal duct

5. Complications Involving the Sinuses

- Surgery
 - o Damage to orbit – the orbit lies lateral to the ethmoid sinus and superior to the maxillary sinus and its contents are at risk during sinus surgery and in severe cases can cause loss of site
 - o Anterior Skull base – lies just above the sphenoid and ethmoid sinuses and can be breached during surgery which can cause a CSF leak and in the worst cases brain damage.
- Rhinosinusitis
 - o Orbit – infective rhinosinusitis can spread to the orbit to cause periorbital sinusitis which can be site threatening
 - o Intracranial infection – infective rhinosinusitis particularly of the frontal sinus can spread intracranially to cause meningitis and intracranial abscess formation

6. Rhinosinusitis

a) Definition

Inflammation of the nose and the paranasal sinuses characterised by two or more symptoms, one of which should be:

- Either nasal blockage/obstruction/congestion or nasal discharge

- (Anterior / posterior nasal drip):
- +/- facial pain/pressure
- +/- reduction or loss of smell

And either:

- Endoscopic signs of - Polyps, mucopurulent discharge, or oedema in middle meatus,

And/or:

- CT changes- Mucosal changes within the osteomeatal complex, or sinuses.

b) Classification

Rhinosinusitis can be classified into **Acute (ARS)** or **Chronic (CRS)**.

- **Acute:**
 - o < 12 weeks, complete resolution of symptoms.
 - o ARS can be divided into viral, and non viral.
- **Chronic:**
 - o 12 weeks, without complete resolution of symptoms.
 - o CRS can be divided into:
 - CRS with nasal polyps**
 - CRS without nasal polyps**

c) Acute Rhinosinusitis

Viral ARS (common cold) is caused mainly by Rhinovirus, and Influenza virus, normally with resolution of symptoms within 5 days.

Non viral ARS is considered to be the persistence of symptoms after 5 days, caused mainly by super added bacterial infections such as *Streptococcus pneumoniae*, *Haemophilus influenzae*, and *Moraxella catarrhalis*.

Allergy and ciliary impairment can predispose to

ARS. Management

- Analgesia if required
- Nasal decongestants
- If persists longer than 5 days consider topical nasal steroids and oral antibiotics

d) CRS with or without nasal polyps

(i) Predisposing factors

- Allergy: Atopy can predispose and is associated with CRS.
- Infections: Bacterial such as *Staph aureus* and *streptococcus pneumoniae*, and fungal infections.
- Ciliary impairment; such as in Cystic fibrosis due to inability of cilia to clear mucus. Nasal polyps are present in about 40% of patients with cystic fibrosis.
- Anatomical abnormalities, such as septal deviation and abnormal uncinate process leads to narrow infundibulum and occlusion of osteomeatal complex.

- Immunocompromised host
- Aspirin hypersensitivity.
- Atmospheric irritants: smoking, dusts, fumes.
- Hormonal: pregnancy and hypothyroidism, nasal congestion is high during pregnancy due to oestrogen and progesterone effect on nasal mucosal vascularity.
- Trauma: Surgical (oroantral fistula), nasal sinus fracture.
- Foreign body in the nose or sinuses
- Swimming and diving

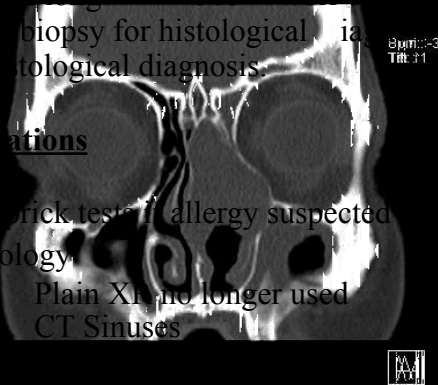
(ii) Nasal polyps

A polyp is simply an abnormal mass which can be found in any part of the body. Nasal polyps are inflammatory polyps. They represent the extreme end of the spectrum seen in CRS. They are normally bilateral and arise from the middle meatus. Diagnosis is from the history and examination. Biopsy for histological confirmation is not necessary. Unilateral polyps require a biopsy for histological diagnosis.

(iii) Investigations

- Skin prick tests if allergy suspected
- Histology

Plain X-ray no longer used
CT Sinuses



Examination

Large numbers of asymptomatic patients have changes in the sinuses on CT scanning

Coronal CT scan showing left sided rhinosinusitis with nasal polyps.

(iii) Management

For the majority of patients there is no cure for CRS and treatment is aimed at improving symptoms.

- Conservative
 - o Avoidance of possible allergens
 - o Nasal douching
- Medical
 - o Antihistamines
 - o Topical nasal steroids
 - o Oral steroids(1 week course) in severe cases
 - o Oral antibiotics
- Surgical treatment
 - o Nasal polypectomy – very high rate of recurrence
 - o Functional Endoscopic Sinus Surgery to improve ventilation/drainage of sinuses
 - o Other procedures to improve nasal airways such as septoplasty, and reduction of inferior turbinates may be considered.

e) Allergy

Allergic rhinitis is an IgE-mediated, type 1 hypersensitivity reaction in the mucous membranes of the nasal airways. The disease is very common, affecting approximately 30% of the Western population and has a strong association with asthma. It can be either seasonal (summer hayfever) or perennial (sometimes with seasonal exacerbations). The commonest allergens include pollens, moulds, house dust mites and animal epithelia. There is a new classification of **Allergic Rhinitis** according to its **Impact on Asthma (ARIA)**, and other health issues. This is based on the **duration**, and **severity** of symptoms:

(i) Duration of symptoms

- Intermittent: symptoms < 4 days per week, and less than 4 weeks
- Persistent: symptoms > 4 days per week, and more than 4 weeks

(ii) Severity of symptoms

- Mild: normal daily activities, and sleep. No troublesome symptoms
- Moderate to severe: Impairment of daily activities and sleep. Troublesome symptoms.

Allergic reaction leads to synthesis and release of arachidonic acid metabolites

(prostaglandin D&leukotrienes) and mast cell degranulation to release histamine. The effect is to increase capillary permeability which leads to congestion, oedema, rhinorrhoea, sneezing and irritation.

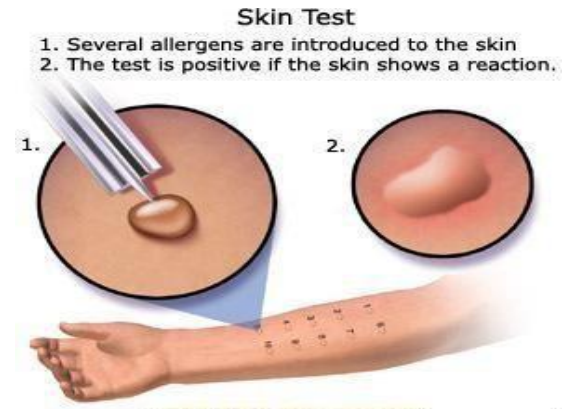
(iii) Investigations

- Skin prick tests (SPT) for specific allergens

- RAST blood tests if SPT not possible

(iv) Treatment

- Conservative
 - o Allergen avoidance
 - o Nasal douching
- Medical
 - o Antihistamines
 - o Topical nasal steroids
- Immunotherapy



7. Peri-orbital cellulitis & Orbital abscess

This is a sight threatening emergency which is more common in children and often results from direct spread of pus from the ethmoid sinus, or from thrombophlebitis of mucosal vessels in any of the sinuses. Patients develop pain followed by oedema of the eyelids and with orbital collection the eye will become proptosed and eye movements will be reduced. There is a risk of blindness as a result of tension and septic necrosis of the optic nerve. Colour blindness is an early sign. A CT scan will confirm the collection and extent of the disease. Treatment with intravenous antibiotics, nasal decongestants, and urgent surgical drainage of any abscess is indicated.

Pharyngeal Infections & Deep Neck Space Infections

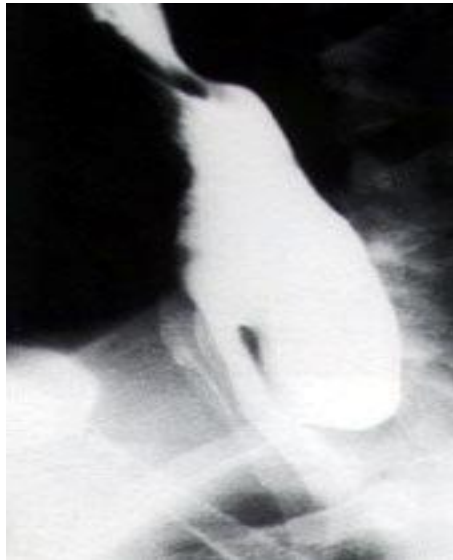
Anatomy of Pharynx and Oral Cavity

Dividing the pharynx and oral cavity into four separate entities allows surgeons to localise lesions and predict their behaviour depending on the surrounding anatomy.

The regions are divided as follows:

- Oral cavity – Extends from the lips to the posterior soft palate.
- Nasopharynx – Superiorly bound by the base of the skull, and inferiorly by an imaginary line at the level of the soft palate
 - o Contents include the adenoids and eustachian tube opening
- Oropharynx – Extends from the level of the soft palate to the superior border of the epiglottis
 - o Contents include the palatine tonsils, anterior and posterior tonsillar pillars
- Hypopharynx – Extends down from the superior border of the epiglottis to the inferior border of the cricoid cartilage. It is posterior to the larynx.

Unlike the rest of the GI tract where the muscular layers are arranged in circular and longitudinal layers, the pharynx has only a circular layer which is formed in principle by four muscles; Superior, middle, and inferior constrictors, and cricopharyngeus. Between inferior constrictor and cricopharyngeus, there is an area deficient of muscle at which herniation may occur. This is known as Killian's dehiscence, and is the site of pharyngeal pouch formation. Pharyngeal pouches typically present with dysphagia, delayed regurgitation of food and sometimes recurrent chest infections from aspirated food.



Barium swallow of pharyngeal pouch

In addition, three pairs of muscles cause elevation and depression of the pharynx - stylopharyngeus, salpingopharyngeus, and palatopharyngeus.

a) Ludwig's Angina

Infection of the space between the floor of the mouth and mylohyoid, most commonly associated with dental infection.

(i) Clinical Features

- Swelling of the floor of the mouth
- Painful mouth
- Protruding tongue
- Airway compromise
- Drooling

(ii) Investigations

- CT neck
- OPG

(iii) Management

1. Secure airway if any concerns
2. IV antibiotics
3. Surgery to drain any collection

(b) Epiglottitis

This is mainly seen in children aged 2 - 6 and is an EMERGENCY.

(ii) Cause

- Normally Haemophilis influenza
- Incidence has reduced with introduction of HIB vaccine

(iii) Presentation– rapidly progressive

- Stridor
- Drooling
- Pyrexia

(iv) Management

- SECURE THE AIRWAY
 - o The patient should not be examined as this may precipitate airway obstruction.
 - o The child should be kept in a calm environment with their parents to avoid undue distress to the child that may also precipitate airway obstruction.
 - o If possible the child should be taken to theatre to be intubated by a paediatric anaesthetist with an ENT surgeon on standby to do a surgical airway if intubation is not possible. If transfer to theatre is not safe they should be intubated in Accident & Emergency where they normally present.
- iv antibiotics
 - o Patients will normally respond after a couple of days of iv antibiotics after which they can be extubated

(c) Tonsillitis

Tonsillitis is a common infection mainly seen in childhood and young adults.

a) Microbiology

(i) Bacterial

- Beta-haemolytic Streptococci
- Staphylococci
- Streptococcus pneumoniae
- Haemophilus influenzae
- Escherichia coli

(ii) Viral

- Rhinovirus
- Adenovirus
- Enterovirus
- Epstein-Barr virus (EBV)

b) Clinical Features

- Pyrexia
- Dysphagia
- Lymphadenopathy
- Odynophagia
- Trismus
- Swollen tonsils with or without exudate
- Otalgia (Referred pain)

c) Management

- Analgesia
- Antibiotics
- Drainage of any peritonsillar abscess
- Tonsillectomy for recurrent tonsillitis

Treatment with amoxicillin should be avoided as this will cause a maculopapular rash in the presence of EBV (glandular fever). Furthermore, patients with EBV should be advised to avoid contact sports for 2-3 months as the virus commonly causes hepatosplenomegaly.

Infection may also be complicated by a peri-tonsillar abscess (quinsy). These normally present with a sore throat that is lateralised to one side and a “hot potato” voice. They may be amenable to drainage under local anaesthetic.

Deep Neck Space Infections

(a) Retropharyngeal abscess

Anterior to the prevertebral fascia, behind the pharynx, is a potential space, the retropharyngeal space, where an abscess may form. This space extends from the base of the skull to the mediastinum.

(i) Clinical Features

- Commonly in young children, commonly after an URTI
- Neck held rigid and upright with reluctance to move
- Systemically unwell
- Airway compromise
- Dysphagia/Odynophagia
- Widening of the retropharyngeal space on lateral X-Ray
- Associated mortality due to airway problems & mediastinitis

(ii) Investigations

- CT Neck

(iii) Management

- Secure airway if any concerns
- IV antibiotics
- Surgery - Incision & drainage

(b) Parapharyngeal Abscesses

The parapharyngeal space is a potential space postero-lateral to the oropharynx and nasopharynx which is divided by the styloid process. Parapharyngeal abscesses may present in a similar manner to peritonsillar abscesses or “quinsy”. There is typically a history of febrile illness, odynophagia, trismus, reduced neck movement and a swelling in the neck around the upper part of the sternocleidomastoid. The parapharyngeal space contains the carotid sheath, and there is therefore the risk of severe complications in this area.

Management

- Secure airway if any concerns
- IV Antibiotics
- Surgical drainage

Head & Neck Cancer (excluding thyroid and salivary gland)

a) Clinical Features/Presentation

- Dysphonia (esp laryngeal malignancy which may cause hoarseness)
- Dysphagia/odynophagia
- Dyspnoea –stridor from narrowing of airway, especially laryngeal tumours
- Neck Mass
- Pain from site of pathology or referred e.g. to ear
- Bleeding from nose or mouth depending on site of primary(rare presentation)
- Nasal blockage –normally unilateral progressive for nasal/nasopharyngeal pathology

b) Histopathology

The majority (90%) are squamous cell carcinomas.

c) Epidemiology

Men are affected twice as much as females.

d) Risk Factors

Alcohol

Tobacco

Beetle nut chewing for oral cavity malignancies

Chinese ethnic origin for nasopharyngeal malignancy

e) Investigations/Staging

The purpose of the investigations in head and neck cancer is to get a histological diagnosis and to stage the disease. From this information a management plan can be formulated and a decision made as to whether any treatment given as part of this management plan will be with a curative intent or for palliation only.

The most commonly used staging system in Head & Neck cancer in the UK is the TNM staging system which is also commonly used in other parts of the body.

T= Tumour size, 1 (small) – 4 (big)

N= Nodal Metastasis, 0 (no nodes) - N3 (big/multiple nodes) M=

Distant metastasis 0 (none) – 1 (present)

The higher the number for each of these components for the TNM system the worse the prognosis. Patients with distant metastases are not curable and should therefore only be treated with palliation in mind. Patients without distant metastases can potentially be cured but as mentioned the higher the T&N number the worse the prognosis and also associated morbidity from treatment so all head and neck cancer patients should have their management discussed at a multidisciplinary meeting with the patient's co-morbidity and personal views in mind.

(i) Investigations of primary tumour site

- Examination under anaesthetic
 - o Also called Panendoscopy or laryngopharyngo-oesophagoscopy in H&N
 - o Purpose – Biopsy for histological diagnosis, assess size of tumour, look for second primary
- CT Neck
 - o Purpose – assess size of tumour and neck node metastasis

(ii) Investigation of neck metastasis

If a patient has a biopsy proven primary squamous cell carcinoma in the head and neck it can be assumed that any palpable cervical lymphadenopathy is secondary to this primary and the CT scan of the neck to assess the primary will give all the information required for staging. If this is not the case any palpable neck masses should be investigated with an ultrasound guided fine needle aspiration (FNA). While blind (i.e. no US) FNA can be done the ultrasound will increase the accuracy of the specimen taken and will give extra information on the mass being sampled.

Sometimes patients with head and neck squamous cell carcinoma may present with neck metastases with no obviously primary even after examination under anaesthetic and imaging. These should be diagnosed with a FNA of the neck mass.

While it is possible to do an open biopsy on these lymph nodes this can lead to seeding of the tumour through the wound onto the skin and end up in a worse prognosis for the patient. Although FNA is very good at diagnosing squamous cell carcinoma it is not so good for other causes of cervical lymphadenopathy such as tuberculosis and lymphoma which often require an open biopsy so in these cases an FNA should be carried out to exclude SCC before an open biopsy is done.

(iii) Investigation of distant metastasis

CT chest are routinely done to check for metastasis in the chest, the commonest site of distant metastasis.

Other investigations are not routinely done unless there is suspicion of other distant metastasis.

f) Management

(i) Palliation

Treatment should be aimed at reducing suffering and in some cases prolonging life. This may involve chemotherapy and radiotherapy and occasionally surgery.

(ii) Curative

Treatment needs to be aimed at the primary site and also any neck metastases. In some head and neck cancers there is a high rate of metastasis to the neck so even if no neck metastasis are detected treatment will still be given to the neck. The possible treatment options are

- Radiotherapy – to the primary site +/- to the neck, +/- chemotherapy
- Surgery
 - o Endoscopic – e.g. laser resection
 - o Open surgery
 - To primary site – e.g. laryngectomy
 - To neck – i.e. neck dissection

Obstructive Sleep Apnoea

Obstructive sleep apnoea and snoring can be considered as a spectrum of disease. Snoring is caused by partial obstruction to the upper airway during sleep from anywhere from the level of the nose to the larynx. In obstructive sleep apnoea there is a complete obstruction of the airway which requires the patient to wake up to alter their position in order to open up the airway again. Repeated apnoeas in a night lead to a poor night's sleep and strain on the cardiovascular system. In children the commonest cause is adenotonsillar hypertrophy whereas in adults it is obesity.

a) Investigation

- BMI
- TFT – ?Hypothyroidism
- CXR – ?Signs of obstructive lung disease
- ECG – ?Signs of right ventricular failure
- Sleep study

b) Treatment

- Advice and lifestyle changes including weight loss
- CPAP – Continuous Positive Airway Pressure (mainstay of OSA treatment)
- Mandibular positioning devices in selected cases
- Surgery –adenotonsillectomy in children, rarely a treatment for adults

SPECIAL SENSES – Clinically

Applied Eye Anatomy

Functional and Applied Anatomy & Physiology of the Eye

Facial injuries may involve the bones of the orbit and the surrounding structures such as the eyeball, nerves, blood vessels and air sinuses. It is important to have knowledge of the eye and its environment. This will require study of the bony orbit, the muscles and neurovascular structures associated with the eye and the structure of the eyeball.

The orbit is a pyramidal cavity in the facial skeleton which contains the eyeball and its associated muscles together with a number of nerves and blood vessels. The periosteum (*periorbita*) lining the bones of the orbit forms the fascial sheath of the eyeball and is continuous at the optic canal and supraorbital fissure with the periosteal layer of the dura mater; over the orbital margin and through the infraorbital fissure, it is continuous with the periosteum covering the external surface of the cranium. Apart of the lateral wall of the orbit, its remaining walls, particularly the medial and inferior, are thin and may fracture as a result of blows and direct trauma.

Eyelids and the lacrimal system

The front of the eye is exposed and the eyelids and lacrimal glands are important protective mechanisms.

The eyelids, which protect the cornea and the eyeball from injury, keep the cornea moist by the covering it with a tear film. On their inner surface, the eyelids are lined by palpebral conjunctiva that is reflected onto the eyeball where it is continuous with the bulbar conjunctiva overlying the anterior surface of the eye.

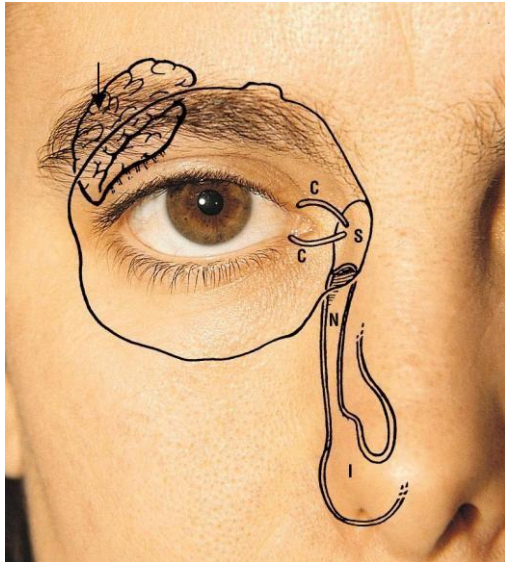
Movement of the eyelids occurs via the orbicularis oculi and levatorpalpebraesuperioris muscles innervated by the facial and oculomotor nerves respectively.

The eyelids are supported by a dense fibrous tissue (tarsal plate).

The tarsal plate contains the tarsal (meibomian) glands, a modified sebaceous glands secreting lipid, which is discussed further below.

The eyelid margin has 2 to 3 rows of eyelashes which are more numerous on the upper eyelid. The tarsal glands open at the lid margin. Modified sweat glands (*glands of Moll*) and sebaceous glands (*glands of Zeiss*) also open at the lid margin.

Obstruction to the tarsal glands can lead to a painless swelling of the eyelid, known as a *chalazion*. A „stye“ or *hordeolum* can also result from acute infection of a gland of Moll, Zeiss or the eyelash follicle. If this is the case it is called an external hordeolum. A swelling involving the tarsal gland is an internal hordeolum (chalazion)



(Diagram above page 68)

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which it passes into the nasopharynx and swallowed.

As there are no anastamotic pathways for drainage of tears, any obstruction to this system will give rise to an overflow of tears in the eye, *epiphora*. Further, stagnant tears may predispose to infection.

Tear film

Tears „wet“ the ocular surface and a tear film is regenerated every time blinking occurs. We need tears to provide a smooth ocular surface for light rays to be refracted uniformly and also provide lubrication to prevent friction between ocular surfaces during lid closure. Further, the tears possess antibacterial properties.

The tear film has 3 components:

Surface lipid layer – secreted by the meibomian (tarsal) glands

Middle aqueous layer – secreted by the lacrimal gland and accessory lacrimal glands

Inner mucus layer – secreted by goblet cells of the conjunctiva and the epithelial cell surface

Dry eyes

The tear film prevents the ocular surface from „drying out“. If this were to occur, the epithelial surface of the cornea and conjunctiva would become damaged. The patient would experience grittiness and discomfort of the eye.

The lipid layer of the tear film plays a very important role. This layer prevents evaporation of the underlying aqueous component. Any deficiency of the tarsal glands e.g. obstruction to their openings, would lead to increased evaporation of tears. Subsequently, the eye becomes „dry“ and there is a risk of damage to the ocular surface and predisposition to infection.

Another way in which „dry eyes“ can occur is that of reduced production of tears. In this case it is important to review the patient for any suggestions of autoimmune disease such as Sjogren's syndrome, rheumatoid arthritis and others. Also, it is important to assess their medications to determine if these could be contributing to reduced tear production e.g. antihistamines.

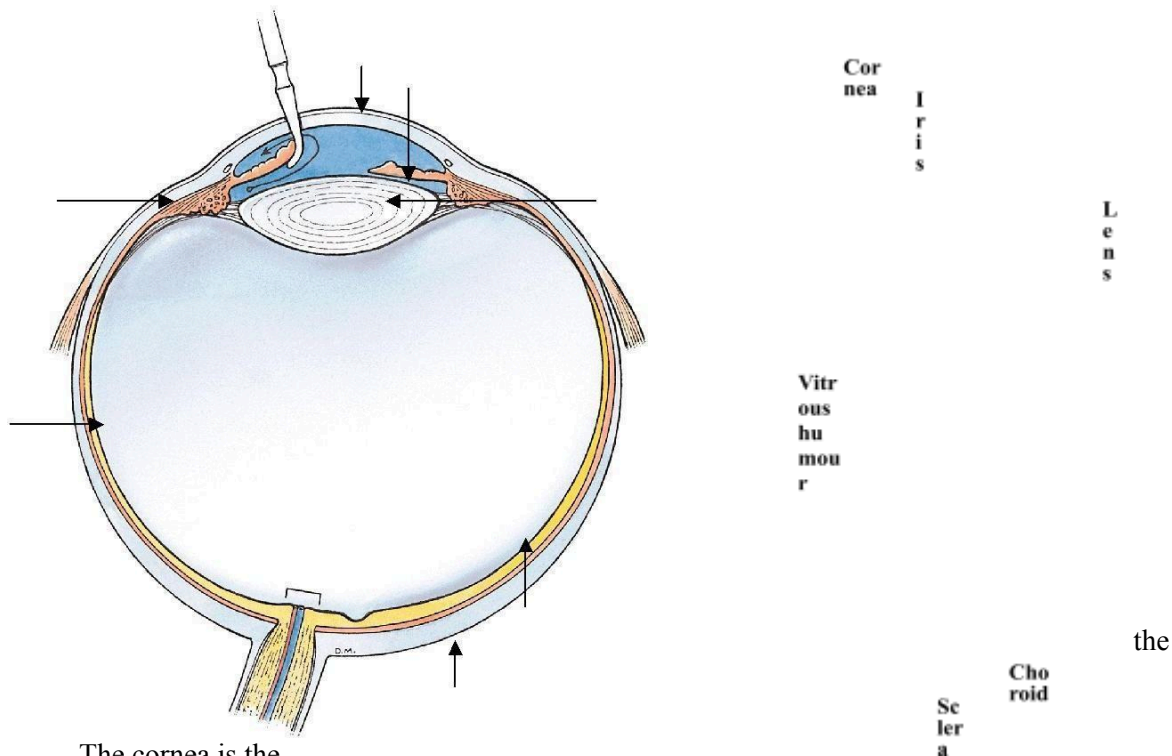
Indirect trauma or injury that displaces orbital contents is called a “blowout fracture”.

Any of the glands in the eyelids may become inflamed and swollen from infection or obstruction of their ducts. When the ducts of the ciliary glands, large sebaceous glands associated with the eyelashes, become obstructed, a painful swelling (stye) forms in the eyelid.

Injury to the facial nerve may result in paralysis of the orbicularis oculi muscle. This prevents the eyelids from closing fully and loss of protective blinking of the eye. As a consequence of this, the cornea becomes dry and is left unprotected from dust and other small particulate material. Irritation of the eyeball results in excessive tear formation.

The Eyeball

The eyeball has three layers. The outer protective layer, which comprises the **sclera** and the **cornea**, is fibrous and provides attachment for the extraocular muscles.



The cornea is the transparent part of the fibrous coat. The choroid, which is a dark membrane between the sclera and the retina, forms the largest vascular layer of the eyeball and terminates anteriorly as the ciliary body. The ciliary body (which is muscular as well as vascular) connects the choroid with the iris.

The eyeball has two chambers. The **anterior chamber** is the space between the cornea and the iris. The **posterior chamber** is the space between the iris and the ciliary body and the **lens**. The ciliary body secretes the **aqueous humour** that fills the chambers of the eye. The iris, which lies on the anterior surface of the lens, is a thin contractile diaphragm with a central aperture (the **pupil**) for transmission of light; two muscles (sphincter and dilator pupillae) control the size of the pupil.

The lens, which lies posterior to the iris, is a transparent biconvex structure enclosed in a capsule and is attached to the ciliary body by the suspensory ligaments. Contraction of the

muscle fibres in the ciliary body changes the shape of the lens. The cavity behind the lens is filled by **vitreous humour** (a transparent jelly-like substance) that supports the lens and holds the retina in place.

The retina, which receives the visual light, consists of a **neural layer** (light-receptive) and a **pigmented layer**. In the posterior part (called the **fundus**) of the eye is a circular depressed area, the **optic disc** where the **optic nerve** enters the eyeball. The optic disc contains nerve fibres and no photoreceptor cells. Just lateral to the optic disc lies a small area (macula lutea) of the retina with photoreceptor cells specialised for acuity of vision. The **fovea centralis**, which is a depression in the centre of the macula, is the area of most acute vision. The retina is supplied by the central artery of the retina and drained by the corresponding vein.

The three layers of the eyeball and their functions are covered in more detail below.

In some diseases/disorders (e.g. thyroid disease) the eyeball protrudes slightly, a condition known as exophthalmos. To accommodate this the eyelids part more than normal so that the whites of the sclera are visible all around the cornea and iris, and just at the sides.

Dirt and other extraneous particles cause corneal abrasions that result in sudden pain and excessive tears. Sharp objects may cause lacerations of the cornea. Injury to the sensory nerve supply (from the ophthalmic division of the trigeminal nerve (5th cranial nerve) to the cornea renders it vulnerable by foreign particles.

During old age, the lens becomes harder and more flattened and these changes slowly reduce their focusing capacity (presbyopia). Some elderly people develop partial or complete opacity of the lens (cataract).

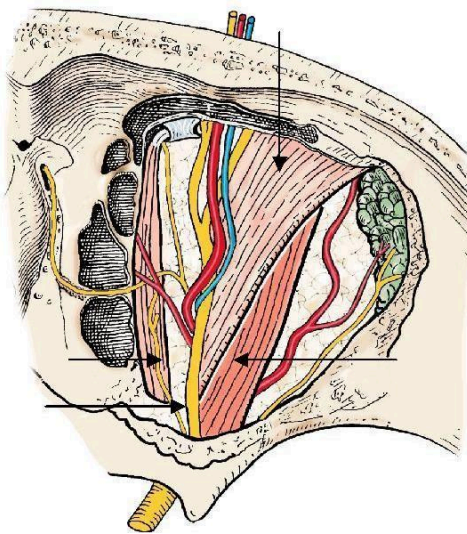
Blunt trauma to the eyeball may result in haemorrhage into the anterior chamber of the eye (hyphema); at first there is red tingeing of the eye followed by accumulation of blood in the chamber.

Since the optic nerve is surrounded by meninges with CSF in the subarachnoid space, increase in the CSF pressure may compress the optic nerve. This in turn compresses the blood vessels supplying the retina, the vein being occluded before the artery. Slow venous return causes oedema of the retina.

The normally depressed optic disc forms a papilla (papilloedema). This is easily seen during ophthalmoscopy. Continued compression of the optic nerve in such conditions may lead to visual imparity.

During embryonic development, the layers of the retina are separated and fuse during the foetal period. Although the pigmented layer is firmly fixed with the choroid, its attachment to the neural layer is not firm. Following a blow to the eye, the two layers may separate causing a detached retina.

The Ocular Muscles, Nerves and Vessels



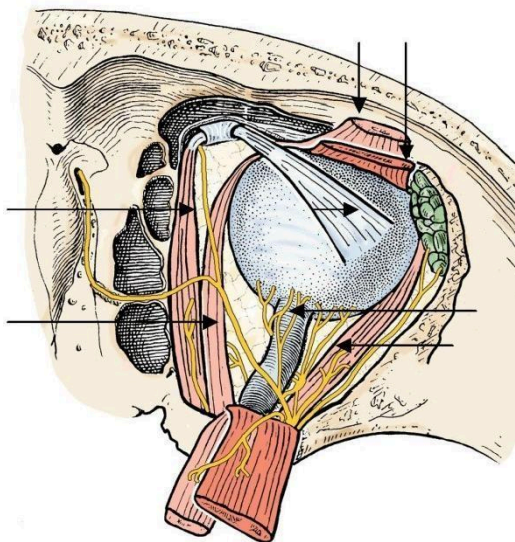
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Complete oculomotor palsy affects most of the ocular muscles, the

levatorpalpebraesuperioris and the sphincter pupillae. The superior eyelid droops (ptosis) and cannot be raised voluntarily because of the unopposed orbicularis oculi supplied by the facial nerve. One or more ocular muscles may be paralysed by head injury or brainstem disease resulting in double vision (diplopia).

Blockage by emboli of branches of the central artery of the retina result in partial or complete loss of vision in the area supplied beyond the blockage. Central retinal vein occlusion presents with variable extent of visual loss.

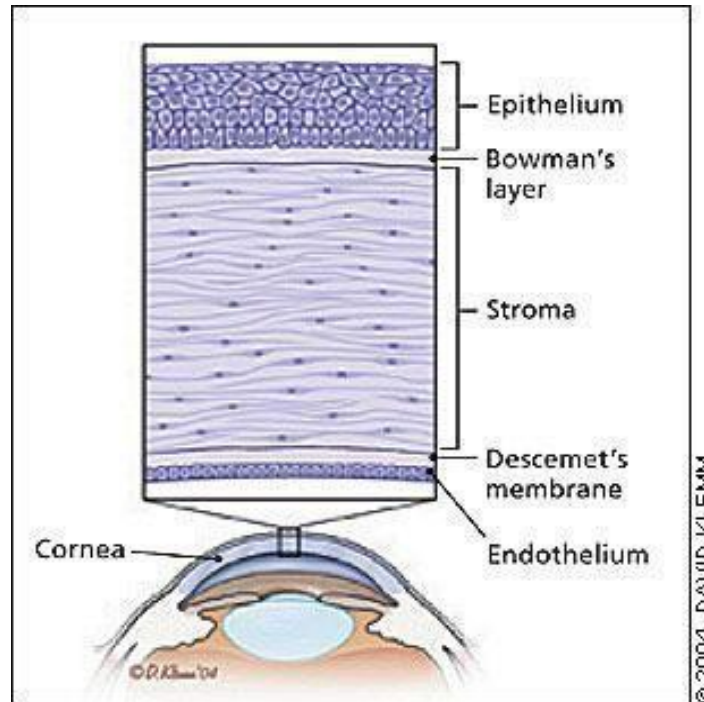
The Cornea, Intra ocular Pressure and the Uveal tract

Cornea

The cornea is a transparent, avascular structure which separates the anterior chamber posteriorly and the tear film anteriorly. It has 3 main roles:

1. Maintaining transparency (so that we have a clear window through which to see)
2. Ocular protection (including the corneal reflex)
3. Refraction of incoming light (along with the overlying tear film)

The cornea is divided into 5 layers: Epithelium, Bowman's membrane, Stroma, Descemet's membrane and Endothelium. We shall briefly discuss the roles of the above structures in achieving the functions of the cornea below.



Epithelium

This is a non-keratinized stratified squamous layer approximately 5-7 cell layers thick. The cells migrate from the basal layer to the surface and are eventually sloughed off into the tear film. When the epithelium is damaged cells must migrate from the basal layer to cover the wound. Cells also migrate from the periphery to the centre of the cornea. When the corneal epithelium is damaged cells on the surface spread out to cover the defect before an increase in mitotic activity at the basal layer replenishes the cellular deficit. The total time for regeneration of the corneal epithelium is quoted as between 3 to 14 days, being longer for central epithelial defects. Corneal stem cells are also located at the periphery of the cornea at the „limbus“; also explaining why central corneal wounds heal slower than more peripherally located defects.

Bowman's Layer

This is an acellular layer and separates the overlying epithelium from the underlying stroma. It is important to recognise as any trauma extending below this level will result in corneal scarring as only the epithelium is regenerated.

Stroma

This connective tissue layer forms the bulk of the cornea. It consists of an arrangement of regularly orientated collagen fibres (type 1 predominantly). The regularity with which these are arranged helps to explain the transparency of the cornea. Keratinocytes (fibroblasts) are found in-between these collagen fibres and help tether them together as well as other functions. Corneal oedema is associated with visible lines in the stroma (striae). These are visible corneal stromal lamellae.

Descemet's membrane

This separates the corneal stroma anteriorly from the endothelium posteriorly. It acts as the basement membrane to the endothelium.

Endothelium

The corneal endothelium's main role is to maintain hydration in order for the cornea to remain transparent. It is a layer of simple squamous endothelium that does not regenerate. Any cellular loss is replaced by changes in shape and size of adjacent cells to cover the deficit. At birth we have 3000-4000 cells/mm² and this steadily falls to 2000 cells/mm² in old age. A level of less than 800 cells/mm² leads to corneal oedema and poor vision. Cellular loss may be accelerated by trauma, surgery and exposure to UV radiation.

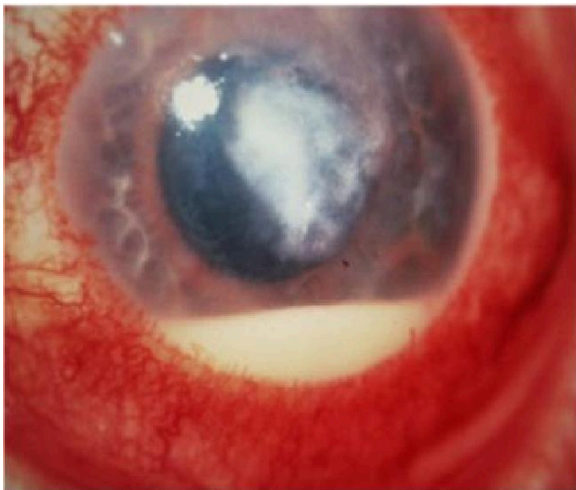
Corneal innervations

The cornea has a rich nerve supply derived mainly from the ophthalmic branch of the trigeminal nerve (via long ciliary nerves). In some cases there is innervation via the maxillary branch which supplies the inferior cornea.

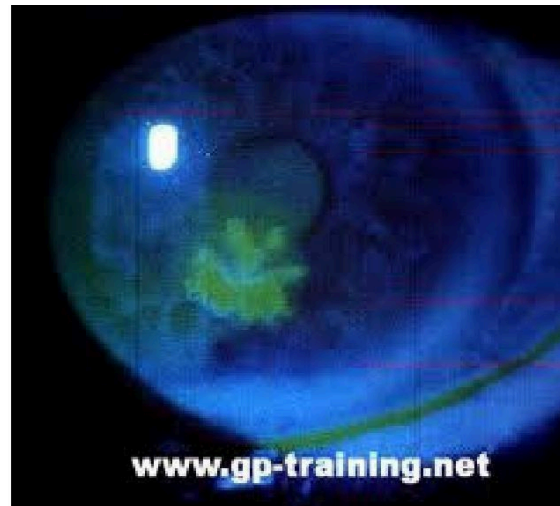
Infective keratitis (kera – cornea)

For infection to occur there is normally the need for disruption to the epithelial surface. Potential triggers for this include contact lens wear, trauma, „dry eyes“, and pre-existing corneal disease amongst others. Also, systemic conditions leading to an immune-compromised state will make infection more likely.

A loss of epithelium exposes free nerve endings causing severe pain. The inflammation also leads to increased peri-corneal vascularity and a red eye. There is an associated watery discharge with epiphora. As the corneal surface and tear film is disrupted along with possible corneal oedema there is a resultant drop in visual acuity. There may also be photophobia and a mucus or muco-purulent discharge. Examination may reveal a white deposit in the cornea (corneal infiltrate) and a collection of pus behind the cornea in the anterior chamber (hypopyon). Looking at the anterior chamber more closely may reveal „cells“ and „flare“ which are leucocytes and protein respectively derived from the leaky iris blood vessels responding to the infection.



a) Image illustrating hypopyon



b) Image illustrating corneal epithelium scarring

Aqueous humour and Intra-Ocular Pressure (IOP)

Aqueous humour is responsible for maintaining an adequate pressure in the eye. IOP is a balance between the rate of production of aqueous and the rate of drainage.

Aqueous humour is produced by the ciliary processes of the ciliary body. It then flows up between the iris and anterior surface of the lens and through the pupil. After passing through the pupil it flows out through the drainage angle of the eye which consists of the trabecular meshwork followed by the Canal of Schlemm. Aqueous then drains into the episcleral vessels and finally into the systemic venous circulation. However, it is also important to note that 10-20% of aqueous is drained via an „Unconventional“ or „Uveoscleral“ route. Here, instead of aqueous passing into the trabecular meshwork it passes into the root of the iris and/or ciliary muscle before draining into the scleral vascular system. This is important as it has pharmacological implications as we shall see below.

IOP is measured by measuring the force needed to flatten the corneal surface using a *tonometer*. The greater the force needed to flatten the cornea, the higher the pressure. The „normal“ range of IOP is 11-21 mmHg. Above 21 is considered to be ocular hypertension. (This measurement does have a correction factor – the corneal thickness

– ask your ophthalmologist about this whilst in clinics during your block.) It is important to note that having high intra-ocular pressure does not equate to a diagnosis of glaucoma. Indeed, a diagnosis of glaucoma can be made with normal IOP. However, high IOP is a significant risk factor for the development of glaucoma.

Autonomic control of IOP

This is via adrenergic receptors. Cholinergic mechanisms have little direct effect on aqueous production.

Alpha 2 receptors – stimulation reduces IOP by reducing aqueous production and may increase uveoscleral outflow

Beta 2 receptors – stimulation increases IOP by increasing aqueous production

Drugs targeting IOP

Once a decision has been reached to lower IOP we can choose from several drugs in order to achieve this. Ideally we would choose a drug that can be used at its lowest concentration and frequency of application has minimal side effects and produces the greatest therapeutic effect. We shall briefly discuss some potential topically used medications below:

CLASS	EXAMPLES AND REGIMEN	MECHANISM OF ACTION
Beta blockers	Timolol, Carteolol OD-BD	Reduce production of aqueous (see above)
Alpha agonists	Apraclonidine, Brimonidine BD	Reduce production and small increase in drainage of aqueous
Prostaglandin analogues	Latanoprost, Bimatoprost ON	Increase uveoscleral outflow of aqueous
Carbonic Anhydrase Inhibitors	Dorzolamide, Brinzolamide TDS	Reduces production of aqueous
Parasympathomimetic	Pilocarpine QDS (when used as monotherapy)	Increases outflow of aqueous by ciliary muscle contraction opening trabecular meshwork

Topical preparations can also be combined, for example, Cosopt – (timolol + dorzolamide).

Lens

The crystalline lens is positioned posterior to the iris and anterior to the vitreous which help to maintain the position of the lens and are aided by the zonules. The zonular fibres (suspensory ligaments) arise from the ciliary body (pars plana) and attach to the surface of the lens.

The lens helps to refract light and although its refractive power is less than that of the cornea; the lens can vary the amount it refracts light by changing its shape – *accommodation*. We require increased accommodation for near tasks. The lens sits in a capsule – a thickened basement membrane that completely envelopes the lens. The lens capsule is responsible for changing the shape of the lens during accommodation. As we age the degree to which the lens is able to accommodate diminishes - a process known as *presbyopia*. The elasticity of the lens together with atrophy of the ciliary muscle contributes to this. We can supplement this drop in accommodative ability by prescribing convex lenses to aid patients with their near work. This is most notable from the age of 40-45 onwards.

The lens is covered with epithelium and the epithelial cells undergo mitosis and eventually form lens fibres. As new fibres are laid down subcapsularly the lens continues to grow throughout life, with the newest layers on the surface and the older layers moving progressively deeper. This also results in the outer lens cortex being soft whereas the central nucleus of the lens is very hard. Lens fibres are tightly packed and the arrangement is highly organised – helping the lens to maintain its transparency.

Cataract

Any opacity (loss of transparency) of the lens is known as cataract and may result from disruption to the lens fibre configuration, capsule or epithelium. The commonest cause for cataract is increasing age. However, cataract can occur in the newborn and it is vital that this is detected as early as possible and referral made to a specialist; in order to enable the best possible vision to be obtained from the affected eye.

Other causes for cataract in younger patients include:

Trauma - direct injury to lens (penetrating), electric shock/lightning, blunt trauma
Drugs – steroids (systemic and topical), amiodarone, allopurinol

Systemic disease - diabetes mellitus, myotonic dystrophy, neurofibromatosis type 2

Morphological types of cataract and symptoms

Along with general symptoms of reduced vision and difficulty with everyday tasks certain cataracts are associated with more specific changes:

NAME	LOCATION	PATIENT SYMPTOMS
Subcapsular	Directly under the lens capsule. Granular or plaque like appearance.	Near vision affected more than distance as opacity at nodal point of eye.
Nuclear	Involving lens nucleus. Yellowish-brown due to deposition of urochrome pigment.	Patient becomes myopic due to increase refractive index of lens. Colours appear less well saturated and more yellow/brown
Cortical	Cortex of lens. Radial spokes in the periphery.	Give rise to astigmatic changes. Patients troubled more in dark when pupil dilates and exposes more of cataract.

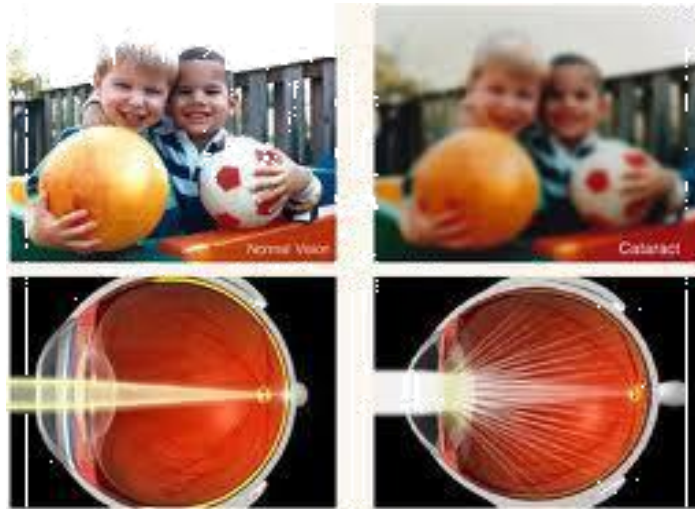


Image illustrating a) Normal vision vs What patients see through a cataract, b) Normal passage of light to the retina, b) Light passing through an opacified lens to the retina

Cataract surgery – Phacoemulsification

During this block you will have the opportunity to see patients with cataracts. It is important that you try to gain an understanding of this patient journey. Phacoemulsification is the most common technique for cataract surgery in the United Kingdom. It is usually performed under local anaesthetic and the patient must be prone. Entry to the eye is made via the limbus (peripheral cornea) using a self-sealing incision. A circular incision is made in the anterior lens capsule – *capsulorhexis*. The cataractous lens is removed using ultrasound and replaced with a plastic lens calculated to correct the patients' refractive error. This is an over-simplification but will hopefully help you when visiting ophthalmic theatres!

The Uveal Tract

The uveal tract consists of the Iris, Ciliary body and the Choroid. These structures are pigmented and vascular. We shall briefly discuss the relevant parts of their anatomy below.

Iris

The iris is a pigmented structure and divides the eye into the anterior and posterior chambers. The iris is anterior to the lens and is attached by the root of the iris at the *irido-corneal* angle where it merges with the ciliary body and trabecular meshwork.

The iris controls the amount of light entering the eye by varying the size of the aperture in the middle of the iris – the pupil. In conditions of low light or sympathetic activation the pupil dilates. This is called *mydriasis* and allows more light into the eye. The dilator pupillae muscle allows the pupil to dilate. It is orientated radially and therefore contraction pulls the pupil outwards and enlarges the pupil diameter.

In conditions of bright light the pupil constricts. This is called *miosis*. The sphincter pupillae allows the pupil to constrict. It encircles the pupillary margin and thus contraction will bring the edges of the pupil closer together resulting in a narrowing of the pupil diameter. It is stimulated via

parasympathetic activation, the fibres of which are carried in the inferior division of the 3rd nerve.

Armed with the above knowledge it should now be simple to determine the effects of the following drugs on the pupil:

Tropicamide – antimuscarinic – mydriasis

Phenylephrine – sympathomimetic –

mydriasisPilocarpine – muscarinic agonist – miosis

With age the dilator pupillae becomes atrophic and the sphincter pupillae more fibrotic, resulting in miosis of the pupil with age.

Ciliary Body

The ciliary body is also a pigmented structure and its primary roles are those of accommodation, aqueous humour production and providing attachment for the lens zonules. The ciliary muscle helps to control accommodation by way of the attached zonules (suspensory ligaments) (see section on lens) which travel up and attach to the equator of the lens. When the ciliary muscle contracts it moves forwards and inwards, thus slackening the tension on the zonules and increasing the power of accommodation.

The ciliary processes are found on the anterior part of the ciliary body. Aqueous humour is produced by the ciliary processes and secreted out into the posterior chamber (see section on Aqueous humour and IOP).

Choroid

The choroid is found covering the posterior part of the eye, extending from the ciliary body to the optic nerve. It divides the retina from the sclera. As with the iris and ciliary body it too is a pigmented structure. The choroid has several important functions:

1. Allows nerves and vessels to reach the anterior eye by passing through the choroid
2. Removes waste product from the outer retina (area closest to choroid)
3. Supplies essential nutrients to the outer half of the retina
4. Absorbs any light passing through the retina, thus preventing it from reflecting back and interfering with vision

The choroid attaches to the retina by way of a basement membrane known as Bruch's membrane. This is important in the pathophysiology of Age Related Macular Degeneration. Moving outwards we then come to the rich capillary bed of the choroid – *choriocapillaris*. It is this layer that supplies the outer half of the retina.

Uveitis

Having defined the uveal tract above it should now be easy to define uveitis – inflammation of the uveal tract. We can anatomically classify uveitis using the following terms:

Anterior uveitis - inflammation affecting the iris (*iritis*) +/- the ciliary body (*iridocyclitis*)

Intermediate uveitis - inflammation of the posterior part of the ciliary body and nearby peripheral retina and choroid

Posterior uveitis – inflammation of the retina and

choroid*Panuveitis*– inflammation of the whole uveal tract

Anterior uveitis

The patient complains of photophobia, pain, reduced vision and a watery eye that may overflow (epiphora). Examination reveals circum-corneal injection, cellular deposits on the corneal endothelium. Examination of the anterior chamber reveals cells and flare (protein), which are derived from leaky iris blood vessels. In the subacute or recurrent cases there may be posterior synechiae – adhesions between the inflamed iris and anterior lens capsule. This is important because if the adhesions became significant they

would obstruct the passage of aqueous humour which flows up between the iris and anterior lens capsule. We therefore prescribe antimuscarinic eye drops (cyclopentolate) to prevent posterior synechiae from occurring. These drops also help with pain derived from spasm of the ciliary muscle as they paralyse the ciliary muscle – *cycloplegia*. Topical steroids are then used to dampen down the inflammation.



Illustration of cells and flare in the anterior chamber

Vitreous

The vitreous cavity fills 2/3 of the volume of the eye. It is positioned behind the lens anteriorly and the retina posteriorly. It is attached to the posterior lens capsule, and to the retina, at several points, posteriorly. It holds the retina in place and supports the lens. It is approximately 99% water and is 3 times more viscous than water. If during cataract surgery, the posterior lens capsule is disrupted, it will lead to disruption of the anterior membrane of the vitreous allowing flow of vitreous into the anterior chamber. With age fluid fills between the potential space between the retina and vitreous. The vitreous may then detach from the retina where it is less weakly bound. A vitreous detachment may predispose to a retinal detachment, especially in areas where vitreous attachments to the retina are firmer.

Retina

The retina is the innermost layer of the eyeball. It has two parts derived embryologically from the optic cup. The first is the inner neural retina and the second is the outer retinal pigment epithelium (RPE). There is a potential space between these two layers - *subretinal space*. The subretinal space between the neural and RPE retina „fuses“ during early foetal life. The RPE is firmly attached to Bruch's membrane (see choroid above). The neural retina is only attached firmly at the optic nerve head and anteriorly at its termination the *ora serrata*. Trauma and other mechanisms can lead to this separation reforming – *Retinal Detachment*. Patients will complain of flashing lights and floaters as well as possible visual field defects due to disruption of afferent nervous pathways.

During your time in this block you may encounter various terms for parts of the retina. It is important that you familiarise yourselves with the following terms – posterior pole, macula/fovea, optic disc and peripheral retina.

Neural Retina

The neural retina is concerned with receiving light stimulus and transforming this into nervous impulses. These impulses travel along ganglion cell axons in the optic nerve, eventually reaching the visual cortex, where they are processed.

There are several neural cell types in the inner retina. However, important cells which we shall discuss briefly are photoreceptors – namely rods and cones. Rods are responsible

for sensing contrast and motion. Rods are not good for detail, „see in black-and-white“ and are also assist seeing in darker environments. Cones sense fine detail and colour vision. Disruption to photoreceptor physiology will lead to specific visual problems. You may wish to read about rod-cone-dystrophies and colour vision anomalies to find out more.

Blood supply

The retina is highly metabolically active and thus requires a good blood supply to meet these demands. The inner 2/3 of the retina are supplied by the central retinal artery whilst the outer 1/3 is supplied by the choroidal blood supply. The central retinal artery divides into 4 and each supplies a segment of the retina. These arteries are end-arteries. Retinal arteries overlie retinal veins and in certain situations eg. Hypertension, characteristic changes can be seen where these vessels cross. The superior and inferior temporal retinal arteries arch over the posterior pole.

It is important to note that the macula has a very dense capillary network, however, the fovea is absent from capillaries – *capillary free zone*. This region is dependant on the underlying choriocapillaris.

Retinal Artery Occlusion

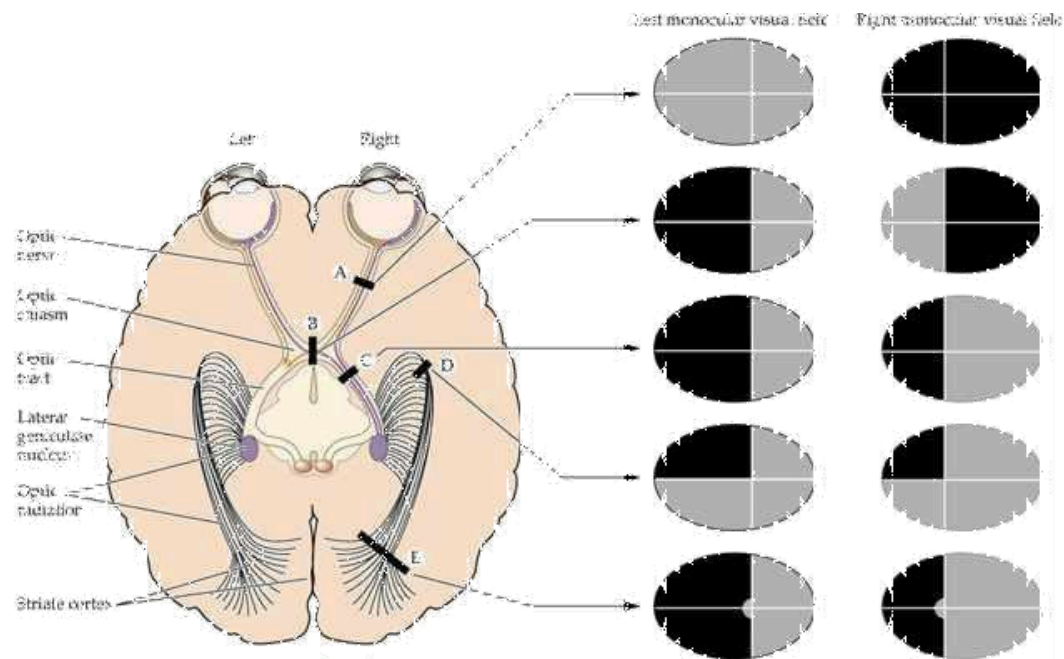
From the above we know that retinal arteries are end-arteries. Therefore, if there is obstruction to an artery there will be insufficient anastomotic supply for tissue survival. In a branch or central retinal artery occlusion the retina becomes pale and oedematous. Changes are irreversible within 1-2 hours. However, in a central retinal artery occlusion the fovea retains a red colour due to being supplied by the choriocapillaris. Patients will complain of painless and sudden loss of vision. Eventually, the inner retina suffers irreversible ischaemia and patients have a remaining visual defect.

Visual pathway

From elemental physics we note that light travels in a straight line. Therefore, light that strikes the temporal retina must come from the nasal visual field and the converse applies for the nasal retina. This simple rule will help when faced with visual field defects and determining locations of lesions causing them, or vice versa.

The temporal and nasal fibres come together at the optic nerve and travel as one to the optic chiasm. Here, the nasal retinal fibres (temporal visual field carrying fibres) decussate (cross over). This means that the right optic tract will carry temporal retinal fibres from the right eye and nasal retinal fibres from the left eye. This puts all the fibres carrying information from one hemi-field together. For the right optic tract this would mean carrying all information from the left half of the visual field. The remainder of the visual pathway maintains this hemi-field organisation of fibres. The optic tract synapses at the Lateral Geniculate Nucleus (LGN). After the LGN the optic radiations arch through the temporal and parietal lobes and terminate at the occipital cortex, (visual cortex). Optic radiations passing through the parietal lobe carry superior retinal fibres from one hemi-field of vision. For example, the right parietal lobe carries superior temporal fibres from the right eye and superior nasal fibres from the left eye. These fibres carry information from the inferior left visual field. Fibres travelling through the temporal lobe are the inferior retinal fibres and would carry information from the superior visual field respectively.

It is important to note that after decussation at the optic chiasm fibres represent one hemi-field of vision. Therefore, any lesion posterior to the chiasm will not, by definition, cross the vertical midline of the visual field. See if you can prove this to yourself!



Imaging showing the visual pathway and possible patterns of visual field loss, taken from: http://www.utdallas.edu/~tres/integ/sen4/8_04.jpg

Pupil reflexes

Afferent pupillary fibres travel with the rest of the retinal fibres but leave the visual pathway just before the LGN. After leaving the visual pathway they synapse with the pretectal nucleus. From here, fibres travel to BOTH Edinger-Westphal nuclei (3rd nerve nucleus). This is important to note as it demonstrates why stimulation of pupillary afferent fibres in one eye causes a bilateral motor pupillary response. Efferent fibres leave the 3rd nerve nucleus, travel to the ciliary ganglion (via the inferior division of the oculomotor nerve) and finally to the sphincter pupillae, causing pupillary constriction.

RAPD

Stimulation of the right afferent pupillary fibres will cause bilateral pupillary constriction. If the right optic nerve (RON) is completely severed we would have no afferent conduction from the right eye. This would mean we get no direct or consensual response from shining a light in the right eye. A consensual pupillary response would remain in the right eye when light is shone into the left eye. This occurs because the efferent system is still intact and can still be innervated by afferent fibres from the left eye innervating both right and left 3rd nerve nuclei. However, what would happen if the RON was damaged incompletely? We would get a sluggish pupillary response when light is shone into the right eye (as there is a reduction in afferent innervation) and a normal right pupillary response when light is shown in the left eye (as the right eye efferent system is intact). This may be subtle and difficult to determine and we can exaggerate this response by swinging the light from the right eye to left eye and back again several times, whilst observing the pupil we are shining light into. If both afferent arms of the pathway are intact we would expect the constriction response to be equal and maintained as we swing our light. If the RON is damaged incompletely and we present the light from the left eye to the right eye, the right eye would not constrict but rather dilate! **This is a rightRAPD.** This is because the afferent information from the left eye which no longer has light shone into it is demanding dilatation rather than constriction. The damaged afferent system of the right eye (into which we are shining light) is overpowered by the intact afferent system of the left eye and instead of constricting, the eye dilates.

Learning with the Ophthalmology Team

Learning Opportunities in the Eye Department

Casualty clinics

These are extremely valuable in allowing you to see patients with acute conditions as well as more minor problems such as corneal foreign body

Paediatric ophthalmology clinics

About 25% of ophthalmology patients are children. Less severe problems such as strabismus or amblyopia in one eye are extremely common, affecting up to 5% of the population. Many of these problems are managed by orthoptists. Even though these children may have a low risk of bilateral visual impairment, the impact on the child and family of the condition and of the treatment may be considerable. Children with other conditions may have severe visual impairment with a major impact on their education and social circumstances

Adult Ophthalmology outpatient clinics

Patients are referred to these clinics by GPs, other consultants especially diabetologists, neurologists and oncologists, as well as directly by optometrists

Preoperative assessment clinics preoperative ward rounds and operating theatre

These offer an opportunity for you to follow the patient journey for common surgical conditions such as cataract.

Argon Laser and Fluorescein Angiography clinics

These offer an opportunity to see patients with diabetic retinopathy and other retinal conditions.

Assessment of patients with ocular disease.

Taking the history from a patient with an eye problem

- Identify the onset and duration of visual symptoms and whether visual function has an impact on the patient's lifestyle.
- **Visual loss:** if present, distinguish between central and peripheral visual loss and acute or chronic
- **Discomfort:** a foreign body sensation indicates corneal irritation e.g. corneal foreign body, foreign body under the upper lid, corneal abrasion, corneal ulcer
- **Double vision:** enquire if monocular (often a rather non-specific symptom), or binocular diplopia (indicates misalignment of the eyes)
- **Photopsia i.e. the sensation of localisable flashing light** is an indicator of retinal traction
- **The ability to localise an abnormal visual sensation** suggests retinal pathology e.g. diabetic retinopathy, age related macular degeneration, retinal detachment etc etc.

Performing an Eye Examination

Check the visual acuity

Check the visual fields with the confrontation test

Observe both eyes, looking specifically for discharge, ptosis and lid lesions.

Examine the corneal reflexes and carry out a cover test to detect any misalignment of the eyes

Examine the pupils and in particular look for any opacity within the normally black pupil, look for any irregularity or distortion of the pupil that could be caused by damage to the pupil or adhesions between the iris and the lens, and check for a relative afferent pupillary defect (RAPD).

Examine each eye with a pen torch looking for generalised or circumcorneal redness, corneal opacity such as a foreign body or abscess, any fluid level of blood or pus in the anterior chamber, and any distortion or irregularity of the pupil.

Use a direct ophthalmoscope to examine the red reflex in each eye to check opacities in the ocular media such as those caused by corneal oedema or other opacity, cataract, or vitreous haemorrhage.

Examine the optic disc, macula and retina with the direct ophthalmoscope, taking care to document both normal and abnormal findings (this is easier when the pupils are dilated).

Management of Eye Disease

Investigations

Visual field testing: An initial assessment of the visual fields can be done by confrontation testing. Precise delineation of the visual fields is done with a perimeter. Modern automated perimeters have statistical methods to monitor deterioration in the visual field over a series of visits and to give an estimate of the reliability of the results. You should be able to distinguish between a field defect suggestive of an ocular problem and one suggestive of a neurological chiasmal or postchiasmal problem.

Fluorescein angiography involves intravenous injection of fluorescein which delineates the retinal vessels. A photographic record is made of the distribution of the dye in the retinal vessels, including any leakage from the retinal vessels over a 15 minute period.

Orthoptic assessment involves detailed measurement of the angle of deviation of the eyes in different directions of gaze as well as assessment of binocular function and the assessment of visual acuity in small children.

Ultrasound, CT and MRI are used to image the eye and orbit.

Ophthalmic conditions and their treatment

External eye and acute disorders of the anterior segment of the eye.

You should be familiar with the common causes of a red eye. By the end of the course you should be able to recognise conjunctivitis, corneal abrasion, corneal foreign body, keratitis, iritis, and orbital cellulitis. You should particularly know about common diseases disorders associated with atopic disease, thyroid disease, urogenital disease, ENT disease and rheumatoid disease. You should know about the treatment of common conditions causing a lump on the eyelid. You should be able to treat simple cases of external eye disease and know when it is appropriate to refer.

Chronic visual loss

You should be able to distinguish the characteristic visual disabilities in patients with glaucoma, cataract, age related macular degeneration, retinitis pigmentosa and hemianopia from cerebrovascular disease. You should understand the distinction between painful acute angle closure glaucoma and chronic simple glaucoma. You should be aware of the systemic side effects of some eye drops for chronic simple glaucoma.

Cataract

You should be able to detect cataract and be able to detect an intraocular lens with a pen torch. You should be aware of the relationship between cataract and ageing and diabetes. You should have some idea of the level of vision at which cataract surgery may be considered and of appropriate expectations from cataract surgery. You should be able to detect congenital cataract and understand the importance of early referral of congenital cataract for the prevention of blindness in childhood.

Diabetic Retinopathy and Age Related Macular Degeneration

You should be able to recognise diabetic retinopathy and understand the implications of your clinical findings. You should have a broad understanding of factors that may contribute to the development of diabetic retinopathy, the scope of treatment and the significance of diabetic retinopathy as a cause of preventable blindness in the UK. You should be able to detect age related macular degeneration and understand in which patients there may be scope for treatment.

Paediatric ophthalmology

You should understand the causes and treatment of amblyopia, and how the commonest causes of treatable visual loss in children are detected and treated.

Ophthalmic manifestations of genetic eye Disease, neuroophthalmology, malignant disease, systemic inflammatory disease and rheumatoid disease

You should have some understanding of the ophthalmic features of the major genetic, neurological, malignant, and systemic inflammatory diseases which may present with eye signs. In particular you should appreciate the wider medical and genetic significance of retinoblastoma, retinitis pigmentosa, optic neuritis, amaurosis fugax, homonymous and bitemporal field defects, and inflammatory eye disease.

Epidemiology

By the end of the course you should have a broad understanding of the impact and causes visual loss in the UK and the extent and cause of preventable blindness worldwide.

Mydriatic and Cycloplegic Drops

Uses:

- ~ To dilate the pupil for visualisation of the retina
- ~ Paralyse accommodation in the management of children with amblyopia ~
- Use in refraction of children for the prescription of glasses

Types:

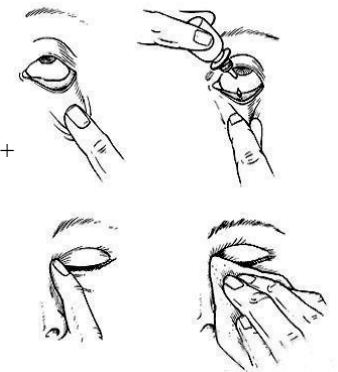
Name	Dose	Mechanism	Length of Action	Contraindications
Atropine	-2 drops of 0.5% or 1% solution	Parasympatholytic- Anticholinergics that blocks the response of the iris sphincter muscles and the accommodative muscles of the ciliary body	-2 weeks	Hypertension Untreated Narrow Angle Glaucoma
Cyclopentolate	-2 drops of 1% or 0.5% solution	Parasympatholytic- Anticholinergics that blocks the response of the iris sphincter muscles and the accommodative muscles of the ciliary body	Effect in 25-75 mins Recovery over 6 – 24 hrs	Untreated Narrow Angle Glaucoma Untreated anatomically Narrow Angle Allergy
Tropicamide	-2 drops of 1% Solution	Parasympatholytic- Anticholinergic that blocks the response of the iris and ciliary muscle.	Effect within 15-20 mins Recovery over 4-8 hrs	Allergy Untreated Narrow Angle Glaucoma
Phenylephrine	-2 drops of 2.5% and 10% solution	Sympathetic agonist - Stimulation of the iris dilation muscle	3-6 hrs	Avoid in children Untreated Narrow Angle Glaucoma

Side Effects:

- ~ Whitening of the eyelids due to vasoconstriction, this will resolve as the drops wear off
- ~ Atropine can cause redness of the face and a warm sensation to the touch, this is not dangerous but a lower dose should be considered ~ All mydriatics will sting the eyes for a few seconds after instillation
- ~ Warn the patient that they cannot drive until the blurring effect of the drops has worn off - this is very important.

Instillation:

- ~ Require : Bottle of drops, Tissues, (Assistant) ~
- Preparation: Hand washing prior to instillation
- ~ Consent: Inform patient of what the procedure will involve, that they cannot drive until the effect have ceased, what the effects will be (dilation + Stinging sensation on application), Check for allergies and contraindications
- ~ Application: Ask the patient to look up and pull the inferior eyelid down to form a well, apply one drop from a vertical position. Allow the patient to blink, wipe away any excess.
- ~ Documentation: Record the type of drops, which eye, how many drops and time of instillation. Time, date and sign.



Fluorescein Drops

Uses:

- ~ Diagnostically to highlight defects in the corneal epithelium
- ~ Can be used as an investigation when measuring the intraocular pressure (tonometry)
- ~ Can be administered combined with a local anaesthetic as a 0.25% solution with oxybuprocaineHCl or proxymetacaineHCl

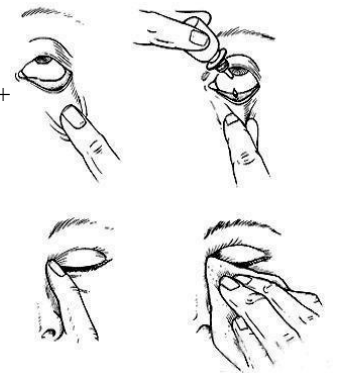
Name	Dose	Mechanism	Length of Action	Contraindications
Fluorescein	-2 drops of 1% or 2% solution	Fluorescein, a precursor of the eosins, temporarily stains any cell it enters therefore marking any damaged areas.	Skin discolouration may last 6-12 hours	Allergy

Side Effects:

- ~ Warn patients about staining of the skin and clothes
- ~ If patient is wearing/will be putting in contact lenses warn that the Fluorescein may discolour them (do not replace soft contact lenses for at least 1 hour)

Instillation:

- ~ Require : Bottle of drops, Tissues, (Assistant) ~
- Prepare: Wash hands
- ~ Consent: Inform patient of what the procedure will involve, that they cannot drive until the effect have ceased, what the effects will be (dilation + Stinging sensation on application), Check for allergies and contraindications
- ~ Application: Ask the patient to look up and pull the inferior eyelid down to form a well, apply one drop from a vertical position. Allow the patient to blink, wipe away any excess.
- ~ Documentation: Record the type of drops, which eye, how many drops and time of instillation. Time, date and sign.



VOCABULARY OF TERMS RELATING TO THE EYE (for reference)

<i>Accommodation</i>	The adjustment of the eye for seeing at different distances, accomplished by changing the shape of the lens through action of the ciliary muscle, thus focusing a clear image on the retina.
<i>Agnosia</i>	Inability to recognise common objects despite an intact visual apparatus.
<i>Albinism</i>	A hereditary deficiency of pigment which affects the eye by causing failure of pigmentation of the retina, iris and choroid. Many different mutations in a number of different genes may cause albinism. In X linked ocular albinism the eye is the only organ affected to a significant extent.
<i>Amaurosis Fugax</i>	Transient recurrent unilateral loss of vision, commonly caused by embolic occlusion of the central retinal artery.
<i>Amblyopia</i>	Reduced vision due to disuse of the eye with no demonstrable organic defect.
<i>Aniridia</i>	Congenital absence of the iris.
<i>Anophthalmos</i>	Absence of a true eyeball.
<i>Anterior Chamber</i>	Space filled with aqueous bounded anteriorly by the cornea and posteriorly by the iris.
<i>Aphakia</i>	Absence of the lens.
<i>Aqueous</i>	Clear, watery fluid that fills the anterior and posterior chambers.
<i>Astigmatism</i>	Refractive error which prevents the light rays from coming to a single focus on the retina because the cornea has a greater curvature in one plane compared to another (i.e. the corneal curvature resembles the surface of a rugby ball).
<i>Binocular Vision</i>	Ability of the eyes to focus on one object and to fuse the two images into one.
<i>Blepharitis</i>	Inflammation of the eyelids.
<i>Blepharospasm</i>	Involuntary spasm of the lids.
<i>Blindness</i>	In the USA, the usual definition of blindness is corrected visual acuity of 6/60 (20/200) or less in the better eye, or a visual field of no more than 20 degrees in the better eye. In the UK blindness is defined as the level of visual impairment such as to prevent any work for which vision is essential. In a patient with normal visual field, an acuity of 3/60 is generally taken to meet the criteria for registration as blind.
<i>Blind Spot</i>	'Blank' area in the visual field, corresponding to the light rays that come to a focus on the optic disc.
<i>Buphthalmos</i>	Large eyeball in infantile glaucoma.
<i>Canaliculus</i>	Small tear drainage tube at the medial end of the upper and lower lid leading from the puncta to the common canaliculus and then to the tear sac.
<i>Canthus</i>	The angle at either end of the eyelid aperture; specified as outer and inner.
<i>Cataract</i>	A lens opacity.
<i>Chalazion</i>	Granulomatous inflammation of a meibomian gland of the eyelid.
<i>Chemosis</i>	Conjunctival swelling from any cause.
<i>Choroid</i>	The vascular middle coat between the retina and sclera.
<i>Ciliary Body</i>	Portion of the uveal tract between the iris and the choroid. It consists of ciliary processes and the ciliary muscle.
<i>Coloboma</i>	Congenital cleft due to the failure of some portion of the eye or lids to complete growth.
<i>Colour Blindness</i>	Diminished ability to perceive differences in colour.
<i>Concave Lens</i>	Lens having the power to diverge rays of light; also known as diverging, reducing, negative, myopic, or minus lens.
<i>Cones and Rods</i>	Retinal photoreceptor cells. Cones are concerned with visual acuity and colour discrimination; rods, with peripheral vision under decreased illumination.
<i>Conjunctiva</i>	Mucous membrane which lines the posterior aspect of the eyelids and the anterior sclera.
<i>Convergence</i>	The process of directing the visual axes of the eyes to a near point.
<i>Convex Lens</i>	Lens having power to converge rays of light and to bring them to a focus; also known as converging, magnifying, hyperopic, or plus lens, denoted by the sign (+).
<i>Cornea</i>	Transparent portion of the outer coat of the eyeball forming the anterior wall of the aqueous chamber.
<i>Corneal Graft (keratoplasty)</i>	Operation in which a section of opaque cornea is replaced with transparent cornea.
<i>Cover Test</i>	A method of determining the presence of misalignment of the eyes.
<i>Crystalline Lens</i>	The lens of the eye.

<i>Cycloplegic</i>	A drug that temporarily puts the ciliary muscle at rest, paralyses accommodation, and dilates the pupil.
<i>Cylindrical Lens</i>	A segment of a cylinder the refractive power of which varies in different meridians, used to correct astigmatism..
<i>Dacryocystitis</i>	Infection of the lacrimal sac.
<i>Dioptre</i>	Unit of measure of strength of refractive power of lenses or of prisms.
<i>Diplopia</i>	Double vision
<i>Ectropion</i>	Turning out of the eyelid so that the conjunctiva lining the inner surface of the lid is exposed to the exterior
<i>Emmetropia</i>	Absence of refractive error; thus an emmetropic individual does not need glasses unless they are old enough to need them for reading i.e. have presbyopia.
<i>Endophthalmitis</i>	Extensive intraocular infection.
<i>Entropion</i>	A turning inward of the eyelid, so that the eyelashes may come into contact with the cornea. This is a common and usually easily treated condition in the elderly in rich countries. (In poor countries, Trachomatous conjunctivitis causes scarring and contraction of the conjunctiva lining the inner surface of the upper lid. and hence entropion. This is a common cause of corneal damage and blindness in Africa.
<i>Enucleation</i>	Complete surgical removal of the eyeball.
<i>Epiphora</i>	Watering from the eye.
<i>Esophoria</i>	A tendency of the eyes to turn inward when binocular reflexes are disrupted by covering one eye (as in the alternate cover test).
<i>Esotropia</i>	An inward deviation of the eyes.
<i>Exenteration</i>	Removal of the entire contents of the orbit, including the eyeball and lids.
<i>Exophoria</i>	A tendency of the eyes to turn outwards when binocular reflexes are disrupted by covering one eye (as in the alternate cover test).
<i>Exophthalmos</i>	Abnormal protrusion of the eyeball which may be caused by dysthyroid eye disease, tumour, infection or bleeding.
<i>Exotropia</i>	A manifest outward deviation of one or both eyes.
<i>Field of Vision</i>	The entire area which can be seen without shifting the gaze.
<i>Floaters</i>	The sensation caused by small opacities moving in the vitreous.
<i>Fovea</i>	Depression in the macula adapted for most acute vision.
<i>Fundus</i>	The posterior of the eye visible through an ophthalmoscope.
<i>Fusion</i>	Co-ordination of the images received by the two eyes into one image.
<i>Glaucoma</i>	Abnormally increased intraocular pressure associated with optic disc damage and visual field loss. Common types of glaucoma are chronic simple open angle glaucoma. In this condition the patient is generally asymptomatic until late when they may present with advanced peripheral visual field loss. The other common type of glaucoma is acute closed angle glaucoma which presents as a red painful eye with blurred vision typically, in an older patient with a history of hypermetropia.
<i>Hemianopia</i>	Blindness of one-half the field of vision of one or both eyes.
<i>Stye (Hordeolum externa)</i>	Infection (usually staphylococcal) originating in an eyelash follicle.
<i>Hypermetropia behind (Hyperopia, farsightedness)</i>	A refractive error in which the focal point of light rays from a distant object is the retina. The patient finds near objects to be blurred. Hypermetropia is usually associated with a small eyeball. This may in later life give rise to crowding at the anterior chamber angle and acute closed angle glaucoma.
<i>Hyphaema</i>	Blood in the anterior chamber.
<i>Hypopyon</i>	Pus in the anterior chamber.
<i>Hypotony</i>	Abnormally soft eye from any cause.
<i>Injection</i>	Congestion of conjunctival blood vessels.
<i>Ishihara</i>	A test for colour vision based on the ability to trace patterns in a series of multicoloured charts. The Ishihara chart is commonly used in clinical practice but was specifically developed to detect X-linked recessive red/green colour deficiency.
<i>Pseudoisochromatic colour Plates</i>	More sophisticated colour vision tests can be used to more precisely define colour vision defects.
<i>Keratoconus</i>	Cone-shaped deformity of the cornea. Initially this can be corrected with contact lenses but in some cases corneal transplantation (keratoplasty) may be required
<i>Keratomalacia</i>	Corneal softening, usually associated with avitaminosis A.
<i>Lacrimal sac</i>	The dilated area at the junction of the nasolacrimal duct and the canaliculi.

<i>Limbus</i>	Junction of the cornea and sclera.
<i>Long-sightedness</i>	See Hypermetropia
<i>Macula Lutea</i>	The small avascular area of the retina surrounding the fovea.
<i>Microphthalmos</i>	Abnormally small eyeball.
<i>Miotic</i>	A drug causing pupillary constriction e.g. acetylcholine or pilocarpine.
<i>Mydriatic</i>	A drug causing pupillary dilation such as atropine, adrenaline or phenylephrine.
<i>Myopia</i> to (<i>Nearsightedness</i>)	Refractive error in which the focal point for light rays from a distant object is anterior the retina. The patient has blurred vision for distance. Myopia is most commonly associated with a large eyeball.
<i>Nystagmus</i>	An involuntary, rapid movement of the eyeball which may be horizontal, vertical, rotary, or mixed.
<i>Ophthalmia Neonatorum</i>	Conjunctivitis in the newborn, classically caused by gonococcus with devastating results. Nowadays, more commonly caused by chlamydial infection.
<i>Optic Atrophy</i>	Optic nerve damage characterised by optic disc pallor.
<i>Dispensing Optician</i>	Health Care professional who fits and sells spectacles prescribed by an optometrist or ophthalmologist.
<i>Optometrist</i> (<i>Ophthalmic Optician</i>)	Health Care professional who examines eyes, and prescribes glasses. Optometrists working in the community have a role in detecting eye disorders and referring to a GP or an ophthalmologist where appropriate. In addition they normally fit and sell glasses and contact lenses they prescribe. Some optometrists work in the hospital eye service supporting the role of ophthalmologists.
<i>Ophthalmologist</i>	Medically qualified doctor who can prescribe glasses and undertakes diagnosis and treatment of eye disorders. There are approximately 1000 consultant ophthalmologists in the UK, almost all of whom undertake eye surgery.
<i>Orthoptist</i>	Health care professional trained to detect, measure and provide care for patients with ocular motility disorders and amblyopia. A large proportion of the work of an orthoptist is involved in the care of children with squint.
<i>Oscillopsia</i>	The subjective illusion of movement of objects that may occur with acquired nystagmus.
<i>Palpebral</i>	Pertaining to the eyelid.
<i>Pannus</i>	Infiltration of the cornea with blood vessels.
<i>Papilloedema</i>	Swelling of the optic disc, commonly associated with raised intracranial pressure.
<i>Photocoagulation</i>	A method of causing usually multiple areas of localised destruction of the retina for treatment of certain types of retinal disorders, e.g. diabetic retinopathy.
<i>Posterior Chamber</i>	Space filled with aqueous anterior to the lens and posterior to the iris.
<i>Presbyopia</i>	Blurred near vision requiring reading glasses, typically evident from the age of 45 years caused by progressive failure with age of the capacity to accommodate
<i>Pterygium</i>	A pathological triangular fold of tissue which extends from the conjunctiva over the cornea.
<i>Ptoxis</i>	Drooping of the eyelid.
<i>Puncta</i>	External orifices of the upper and lower canaliculi.
<i>Refractive Error</i> (<i>ametropia</i>)	A defect (such as myopia, hypermetropia or astigmatism) that prevents light rays from being brought to a single focus on the retina.
<i>Retinal Detachment</i>	A separation of the retina from the choroid. Patients with retinal detachment (which is more common in myopic individuals) typically complain of flashing lights (photopsia) and floaters followed by a black curtain spreading across the field of vision. Prompt surgical intervention is usually indicated.
<i>Retinitis Pigmentosa</i>	A hereditary degeneration of the retina.
<i>Retinoscope</i>	An instrument designed for the objective measurement of refraction.
<i>Sclera</i>	The white part of the eye - a tough covering which, with the cornea, forms the external protective coat of the eye.
<i>Scotoma</i>	A blind or partially blind area in the visual field.
<i>Short-sightedness</i>	see Myopia
<i>Slitlamp</i>	A microscope providing focal illumination for examination of the eye
<i>Snellen Chart</i>	Used for testing central visual acuity. It consists of lines of letters or numbers, in graded sizes so that a person with 'standard vision' able to resolve two limbs of a letter "E" subtending one minute of arc is able to read the 6 line at 6 metres. A patient able to read the top "60" line at 6 metres has an acuity of 6/60. American Snellen charts are calibrated in feet. Hence the term 20/20 vision which is equivalent to an acuity of 6/6.

<i>Strabismus (squint, tropia)</i>	Misalignment of the eyes.
<i>Stye</i>	Infection (usually staphylococcal) of an eyelash follicle
<i>Sympathetic Ophthalmia</i>	Inflammation in one eye following traumatic inflammation in the fellow eye.
<i>Synechia</i>	Adhesion of the iris to cornea (anterior synechiae) or lens (posterior synechiae) Iritis is the usual cause of posterior synechiae. In iritis associated with Juvenile Chronic arthritis the only indicator of severe sight threatening eye disease may be irregularity of the pupil caused by posterior synechiae.
<i>Trachoma</i>	Serious infectious keratoconjunctivitis due to repeated chlamydial infection in hot dry climates where flies are prevalent. Worldwide this is one of the commonest causes of blindness. see <i>Entropion</i>
<i>Uvea (uveal tract)</i>	The iris, ciliary body and choroid.
<i>Uveitis</i>	Inflammation of one or all portions of the uveal tract.
<i>Visual Acuity</i>	Measure of detailed central vision used for reading face recognition etc (see also Snellen chart)
<i>Vitreous</i>	Transparent, colourless mass of soft, gelatinous material filling the eyeball behind the lens.
<i>Zonule</i>	The numerous fine tissue strands which stretch from the ciliary processes to the lens equator (360 degrees) and hold the lens in place.

SS BLOCK

END OF BLOCK ASSESSMENT- What to do

Each student should complete the FOUR parts.

- 1. PART ONE: Completed the Attendance paper (fully signed or stamped)**
This is a weekly-assessed paper including the TWO daily activities (morning activity and afternoon activity). This checked by block secretary and block lead on a weekly basis.
- 2. PART TWO: SS BLOCK WORKBOOK**
 - A. Presenting THREE tutorial (ENT, OPHTHALMOLOGY, & DERMATOLOGY)**
 - B. Performing ALL core procedural skills.**
 - C. Attending ALL common surgical procedures (10)**
- 3. PART FOUR: Performed well in the EMQ paper at end of the block.**

SS BLOCK ATTENDANCE PAPER

This paper should be signed or stamped each day twice, by every student. A total of 8 signed activities should be confirmed by the block lead and secretary at the end of each week except the induction week.

Student Name:

DAY	MORNING ACTIVITY	AFTERNOON ACTIVITY
SATURDAY		
SUNDAY		
MONDAY		
TUESDAY		
WEDNESDAY		

BLOCK LEAD SIGNATURE:

SECTION TWO (A): TUTORIAL REQUIRMENTS

STUDENT NAME:

ENT

Seminar/ Tutorial title:

SCORE /100

SENIOR NAME & SIGNATURE

STUDENT NAME:
OPHTHALMOLOGY

SCORE /100

SENIOR NAME & SIGNATURE

**STUDENT
NAME:
DERMATOLOG
Y
Seminar/ Tutorial
title:**

SCORE /100

SENIOR NAME & SIGNATURE

SECTION TWO (B): CORE PROCEDURAL SKILLS

Each student should masterly perform the **ALL** of the following core skills. Firstly it is worthy to watch it done and then perform it under supervision. Each skill needs to be signed separately with **name of senior clearly written**.

ENT:

<u>EAR</u>	Senior's Name	Senior's signature
1. otoscopy		
2. Identify the eardrum		
3. Understanding the tuning fork tests		
4. Identify an abnormal eardrum, including glue ear and perforation of the eardrum		
5. Understand an audiogram and a tympanogram		

<u>NOSE</u>	Senior's Name	Senior's signature
1. Performing an anterior rhinoscopy using a nasal speculum and headlights		
2. Assess epistaxis and various nasal packing devices		

<u>THROAT</u>	Senior's Name	Senior's signature
1. Performing an anterior rhinoscopy using a nasal speculum and headlights		
Performing an examination of the oral cavity and oral pharynx using a headlight and tongue depressors 3. Ability to explain to patients the use of rigid and flexible endoscopy under local anaesthetic		

OPHTHALMOLOGY

<u>EYE</u>	Senior's Name	Senior's signature	
1. Test and record visual acuity in adults and children using a Snellen Chart			
2. Examine the external eye with a pen torch and be able to recognise conjunctivitis, corneal abrasion, corneal foreign body, corneal abscess, blood in the anterior chamber, pus in the anterior chamber and a distorted pupil following trauma or iritis			
3. Assess a patient for presence of a squint by means of the corneal reflexes and cover testing			
4. Examine the pupils including performing the swinging flash light test for a relative afferent defect			
5. Examine the fundus with a direct ophthalmoscope and be able to detect gross abnormalities such as severe diabetic retinopathy, optic atrophy, papilloedema, and age related macular degeneration.			
6. Use safely mydriatic and fluorescein diagnostic drops			
7. Examine visual fields by confrontation			
8. Examine the ocular media of both adults and children using the coaxial illumination from an ophthalmoscope to see the red reflex, and using a torch; be able to recognise the abnormalities characteristic of cataract, vitreous haemorrhage and malignant disease of the eye			

DERMATOLOGY SKILLS

[illegible]

SECTION TWO (C): ATTENDING COMMON SURGICAL PROCEDURES

procedure	Type	Senior name	Signature
TOSILLECTOMY			
ADENOID SURGERY			
SINUS SURGERY			
DL			
MASTOIDECTOMY			
CATARACT Sx			
GLAUCOMA Sx			